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A sequential interpretation of conditional belief

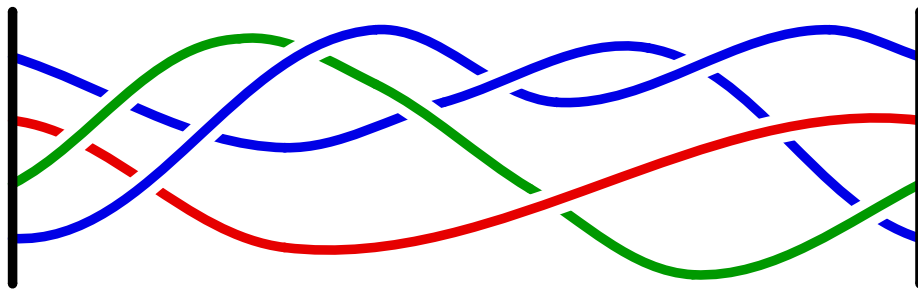
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Abstract

Fagin and Halpern [2] proved that FH conditional belief is itself a belief function. Their proof, however, is quite involved and makes it difficult to understand why this has to be the case. In this thesis, we compare their proof to an alternative proof given by Jaffray [3]. Throughout this comparison, we look not only at how the proofs work, but also at whether their mathematical constructions can be interpreted epistemically. Jaffray's construction involves an auxiliary function that assigns weights to ordered sequences. By interpreting the recursive assignment of these weights through successive changes in plausibility, we find a sequential structure underlying conditional mass. This structure is hard to discern once the sequence weights are aggregated into the final conditioned basic belief assignment. This interpretation of changes in plausibility leads to an adapted version of Jaffray's proof.

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Contents

1. Introduction	5
2. Foundations of conditional belief	7
2.1. Probability on finite outcome spaces	7
2.2. Conditional probability	8
2.3. Incomplete information	10
2.4. Basic belief assignment, belief, and plausibility	11
2.5. Equivalent representations of belief	13
2.6. Design choices in defining conditional belief	14
2.7. Fagin-Halpern conditioning	15
2.8. Dempster conditioning	17
2.9. Central theorem	20
3. Fagin and Halpern	21
3.1. Proof outline and defining focal sets	21
3.2. Reverse-engineering bba through examples	22
3.3. Partial proof by Fagin and Halpern	26
3.4. Limitations of algebraic approach	29
4. Jaffray	30
4.1. From strings to ordered sequences	30
4.2. Jaffray's auxiliary function	32
4.3. Proof by Jaffray	34
4.4. Worked example	38
5. The sequential structure of conditional mass	42
5.1. Noncommutativity and ordered mass	42
5.2. Comparing operations	43
5.3. Mutually exclusive adaptation	46
5.4. General adaptation	51
5.5. Revisiting worked example	53
6. Interpretation and conclusion	56
6.1. The epistemic role of noncommutativity	56
6.2. The sequence-based structure of conditional mass	57
6.3. Conclusion and limitations	58
A. Fagin and Halpern continued	61

1. Introduction

A probability is a mathematical tool for assessing how likely an event is to occur, and just like all tools, it is designed for a specific purpose. Classical probability is designed for settings where each possible outcome has an assigned probability. This can be limiting when describing certain epistemic processes. Suppose, for example, that we combine meteorological observations to assess the chance of rain. Some of the observations may point in conflicting directions. Perhaps the humidity indicates that it will rain, while the barometric pressure indicates that it will not. Aggregated, the model may have 50% of the evidence supporting rain, while having 20% supporting no rain. The remaining 30% then represents unresolved information, and so we may wish to state that “the chance of rain is between 50% and 80%”. Classical probability theory does not allow for $P(\text{Rain}) + P(\text{No rain}) = 0.7$, because it satisfies the rule of complement. This rule states that the probability of an event and the probability of its complement must add up to one. Here we see that probabilities cannot be assigned independently; the value of one constrains the values of others. A complete assignment of probabilities that satisfies such consistency rules is called a probability measure. Using a probability measure is natural when the available information justifies assigning precise probabilities to the outcomes, but less so when part of that information remains ambiguous.

To model such unresolved information, we turn to the theory of belief functions. Rather than forcing every piece of evidence to support one individual outcome, this theory allows evidence to support a set of possible outcomes. So in the rain example, we can assign the remaining 30% of evidential mass to the set containing both outcomes, “rain or no rain”. In this way, the theory of belief functions preserves information that would be lost when forcing the evidence into a probability measure.

This thesis revolves around the operation of conditioning, which is the mechanism by which an epistemic model incorporates new information. For probability measures, conditioning is a natural operation that produces a new probability measure. For belief functions, this is less straightforward. Since part of the available evidence may support multiple outcomes at once, learning that some outcomes are no longer possible raises the question of how this unresolved evidence should be treated. This is as much a mathematical issue as it is an epistemic one. Different answers to this question lead to different definitions of conditional belief. This thesis focuses on the definition adopted by Fagin and Halpern, which we will refer to as FH conditioning.

Just as a conditioned probability measure is again a probability measure, Fagin and Halpern proved that FH conditional belief is itself a belief function [2]. This means that after incorporating new information, the resulting model still has the mathematical properties required of a belief function. Meester and Slooten emphasize the importance of this result, but state: “The proof of this fact, however, is surprisingly involved and

falls outside the scope of this book” [4]. This observation marks the point of departure for this thesis. Why does an operation that appears so natural require such an involved proof?

In this thesis we aim to understand how and why FH conditioning produces a belief function. The theorem itself is already proven, so the contribution of this thesis lies in analysis of the proof. For this we will compare the proof by Fagin and Halpern to another one by Jaffray [3]. We analyze how the proofs work, and investigate what their constructions reveal about FH conditioning as an epistemic tool. At first, comparing the two proofs seems to suggest that the order in which evidence arrives matters for the construction of conditional belief. This intuition turns out to be more nuanced, but investigating it does reveal a sequential structure in Jaffray’s proof and leads to a new interpretation of his construction.

This thesis builds on the theory of belief functions developed by Dempster and Shafer [1, 6]. Fagin and Halpern adopted the lower-envelope definition of conditional belief, derived its closed-form expression, and proved that the resulting function is itself a belief function [2]. Jaffray later provided an alternative proof through an auxiliary function defined on ordered sequences [3]. For the mathematical foundations, this thesis follows the presentation by Meester and Slooten [4, Sections 13.1 and 13.2]. Pearl’s criticism of the noncommutativity of FH conditioning motivates part of the later discussion on the epistemic role of order[5].

The remainder of this thesis is structured as follows. Chapter 2 introduces the mathematical foundations of belief functions and conditional belief, ending with the central theorem. Chapter 3 presents the proof by Fagin and Halpern, while Chapter 4 presents the proof by Jaffray. Chapter 5 then investigates the sequential structure suggested by the comparison and develops an adapted version of Jaffray’s proof based on this interpretation. Finally, Chapter 6 reflects on the epistemic implications of the mathematical results and presents the conclusions and limitations of this thesis.

2. Foundations of conditional belief

The exploration of conditional belief begins at its foundations. This chapter first introduces some concepts from classical probability that are relevant to belief functions. We then introduce belief functions by explaining the problem they are designed to address, followed by their formal definition and the definitions of the related concepts of basic belief assignment and plausibility. Next, we introduce conditional belief as an epistemic tool designed for a specific purpose. We emphasize that conditioning a belief function is a mathematical operation intended to model a real-world epistemic process. To illustrate this, we also consider a few alternative choices that could have been made in its definition. Such choices will not be assessed solely on mathematical grounds, but also by how well they reflect the underlying epistemic process. We end the chapter by presenting the central theorem of this thesis.

2.1. Probability on finite outcome spaces

Describing an uncertain event begins with identifying its possible outcomes. Throughout this thesis, we assume that these outcomes form a finite set, so that all relevant events and assignments can be represented explicitly. We denote this outcome space by Ω . Any subset $X \subseteq \Omega$ is called an event. For example, a roll of a fair six-sided die has outcome space $\Omega = \{1, 2, 3, 4, 5, 6\}$. In that case, the number of outcomes is finite ($|\Omega| = 6 < \infty$), so this is an example of a finite outcome space. Probabilistic experiments can also have infinite outcome spaces. For example, the number of coin tosses required until heads first appears has outcome space $\Omega = \{1, 2, 3, \dots\}$. Such spaces fall outside the scope of this thesis.

One intuitive interpretation of probability is given by the frequentist perspective. Suppose that we throw the fair die repeatedly, and let $N_n(6)$ denote the number of times outcome 6 occurs among the first n throws. The fraction

$$\frac{N_n(6)}{n}$$

is called the relative frequency of outcome 6 after n throws. The frequentist perspective interprets the statement $P(\{6\}) = 1/6$ as saying that this relative frequency approaches $1/6$ as the number of throws increases. The frequentist perspective is a useful way to interpret probabilities. However, a more rigorous definition for a mathematical probability is provided by the concept of a probability measure.

For a given finite outcome space Ω , a probability measure assigns a probability to every event $X \subseteq \Omega$. For a fair six-sided die, for example, it assigns to the event of throwing

a 1, 2, or 5 the probability $1/2$; $P(\{1, 2, 5\}) = 1/2$. Hence, a probability measure is a function that maps any subset of Ω to a real number in the interval $[0, 1]$. We denote the set of all subsets of Ω by $2^\Omega := \{X : X \subseteq \Omega\}$, this notation reflects the fact that Ω contains exactly $2^{|\Omega|}$ subsets.

We now give the formal definition, adapted to finite outcome spaces.

Definition 2.1. Let Ω be a finite outcome space. A function $P : 2^\Omega \rightarrow [0, 1]$ is a *probability measure* if it satisfies the following conditions:

1. $P(\Omega) = 1$.
2. $P(X \cup Y) = P(X) + P(Y)$ for any two disjoint subsets $X, Y \subseteq \Omega$ (meaning that $X \cap Y = \emptyset$).

The connection between a probability measure and some probabilistic experiment still grants further explanation. The presented definition allows us to verify whether a given function from 2^Ω to $[0, 1]$ constitutes a valid probability measure. For a die with $\Omega = \{1, 2, 3, 4, 5, 6\}$, there are many probabilistic experiments we can think of, each of which has its own corresponding probability measure. For example, we could be interested in the last thrown side when throwing a die until the sum of the throws surpasses 10. This specific probabilistic experiment has some valid probability measure that exactly calculates the right probability for each subset of Ω . We can however also think of some new function without having a probabilistic experiment in mind, and still test whether it is a valid probability measure.

Note that the rule of complement mentioned in the introduction, $P(X) + P(X^c) = 1$, follows directly from these two conditions.

2.2. Conditional probability

Suppose that for a given probabilistic experiment we receive some evidence by which we know that some of the outcomes cannot occur. That is, we assume that the outcome lies in some event $E \subseteq \Omega$. Here we assume $P(E) > 0$, this is required so that conditioning on E is well-defined, and it makes sense epistemically since otherwise the event E cannot occur. For any subset $X \subseteq \Omega$ we can then calculate the probability of X given that E occurs, using the definition of conditional probability.

Definition 2.2. Suppose $X, E \subseteq \Omega$ with $P(E) > 0$. Then the probability of X given E is defined by

$$P(X \mid E) = \frac{P(X \cap E)}{P(E)}.$$

To explain a conditional probability we return to the frequentist perspective. In this interpretation, the conditional probability of X given E is the relative frequency of X occurring within the subsequence of trials determined by the occurrence of E . Suppose the experiment is repeated independently under the same probability measure. Let x_i

denote the outcome of the i th experiment, and let $1_A(x_i)$ equal 1 when $x_i \in A$ and 0 otherwise. Among the first n experiments, the numerator $\sum_{i=1}^n 1_{X \cap E}(x_i)$ counts how often both X and E occur, while the denominator $\sum_{i=1}^n 1_E(x_i)$ counts how often E occurs. Their ratio is therefore the relative frequency of X within the trials where E occurred. As the number of experiments increases, this ratio approaches the conditional probability almost surely.

$$P(X | E) = \lim_{n \rightarrow \infty} \frac{\sum_{i=1}^n 1_{X \cap E}(x_i)}{\sum_{i=1}^n 1_E(x_i)}.$$

Consider again the fair six-sided die. Let $X = \{2, 4, 6\}$ be the event that the outcome is even, and suppose we learn that the outcome lies in $E = \{4, 5, 6\}$. Since $X \cap E = \{4, 6\}$, we find

$$P(X | E) = \frac{P(\{4, 6\})}{P(\{4, 5, 6\})} = \frac{2/6}{3/6} = \frac{2}{3}.$$

Thus, after learning that E occurred, only the three outcomes in E remain relevant, two of which are contained in X .

Next we show that conditioning a probability measure results in another probability measure. Let Ω be a finite outcome space, and $P : 2^\Omega \rightarrow [0, 1]$ be a probability measure. Furthermore, let $E \subseteq \Omega$ be an event such that $P(E) > 0$. We claim that the function $P(\cdot | E)$ is again a probability measure. We can confirm this claim by checking the two conditions as follows.

1. First, recall that P is a valid probability measure, hence $P(\Omega) = 1$. Furthermore, by assumption we have $P(E) > 0$. It then follows that

$$P(\Omega | E) = \frac{P(\Omega \cap E)}{P(E)} = \frac{P(E)}{P(E)} = 1.$$

This confirms the first condition.

2. Second, fix any two disjoint subsets $X, Y \subseteq \Omega$. Since X and Y are disjoint, the sets $X \cap E$ and $Y \cap E$ are also disjoint. Hence, since P upholds the second condition for being a probability measure, we know that $P((X \cap E) \cup (Y \cap E)) = P(X \cap E) + P(Y \cap E)$. It then follows that

$$\begin{aligned} P(X \cup Y | E) &= \frac{P((X \cup Y) \cap E)}{P(E)} \\ &= \frac{P((X \cap E) \cup (Y \cap E))}{P(E)} \\ &= \frac{P(X \cap E) + P(Y \cap E)}{P(E)} \\ &= P(X | E) + P(Y | E). \end{aligned}$$

This confirms the second condition.

Thus, having checked both conditions, we have shown that the function $P(\cdot \mid E)$ is a probability measure. Since a conditioned probability measure is again a probability measure, we can condition on one event after the other. Suppose we successively receive new evidence, first some event E_1 and subsequently E_2 , where $P(E_1 \cap E_2) > 0$. Let $P_{E_1} = P(\cdot \mid E_1)$ denote the probability measure obtained after conditioning on E_1 . Conditioning this measure on E_2 gives

$$\begin{aligned} P_{E_1}(X \mid E_2) &= \frac{P_{E_1}(X \cap E_2)}{P_{E_1}(E_2)} \\ &= \frac{P(X \cap E_1 \cap E_2)/P(E_1)}{P(E_1 \cap E_2)/P(E_1)} \\ &= \frac{P(X \cap E_1 \cap E_2)}{P(E_1 \cap E_2)} \\ &= P(X \mid E_1 \cap E_2). \end{aligned}$$

Interchanging E_1 and E_2 gives the same expression. Hence,

$$P_{E_1}(X \mid E_2) = P_{E_2}(X \mid E_1) = P(X \mid E_1 \cap E_2).$$

Thus, for classical probability, successive conditioning is independent of the order in which the evidence is received. Furthermore, conditioning on one event then on another is equivalent to conditioning directly on their intersection. Whether this property also holds for belief functions will be a central focus of this thesis.

2.3. Incomplete information

A probability measure is a function from 2^Ω to $[0, 1]$, this means that it must assign a probability to each event. This aspect can be restrictive when describing real world scenarios where not all required information is present. Let us illustrate this through an example.

Example 2.1. Suppose we throw an unbalanced three-sided die with outcome space $\Omega = \{1, 2, 3\}$. Suppose that some observations are imperfect, so that the number on top cannot always be identified. Occasionally, we can only see a number on the side, allowing us to exclude one outcome. From the frequentist perspective, suppose that repeated observations give us the following relative frequencies for reports:

- 0.5 for the report $\{1\}$;
- 0.3 for the report $\{3\}$;
- 0.2 for the ambiguous report $\{2, 3\}$, where we only know that the outcome is not 1.

Let us assume now that the experiment is governed by some underlying probability measure, and that every report contains the true outcome. Given this setting there are some probabilities we can confirm, like $P(\{2, 3\}) = 0.5$. But to be able to get a complete probability measure for this experiment we have to specify the value of $P(\{2\})$. Hence, we cannot determine the underlying probability measure completely. What we can do is construct possible measures that would be consistent with the available information. To construct such a complete probability measure, the ambiguous mass must be distributed between the outcomes 2 and 3. At one extreme, all ambiguous mass is assigned to outcome 3, giving $P(\{1\}) = 0.5$, $P(\{2\}) = 0$, and $P(\{3\}) = 0.5$. At the other extreme, all ambiguous mass is assigned to outcome 2, giving $P(\{1\}) = 0.5$, $P(\{2\}) = 0.2$, and $P(\{3\}) = 0.3$. Any distribution between these extremes is also consistent with the available information.

What we thus described is a collection of probability measures, each of which is consistent with the available information. We denote this collection of consistent probability measures by $\mathcal{P}$. In this example we can explicitly define $\mathcal{P}$ by parametrizing its probability measures with a variable α . Namely, let $\mathcal{L}$ denote the set of all probability measures on Ω . Then

$$\mathcal{P} := \left\{ P_\alpha \in \mathcal{L} \mid \alpha \in [0, 0.2], \begin{array}{l} P_\alpha(\{1\}) = 0.5 \\ P_\alpha(\{2\}) = \alpha \\ P_\alpha(\{3\}) = 0.5 - \alpha \end{array} \right\}.$$

In the Dempster-Shafer theory of evidence, ambiguity is expressed through the concepts of belief and plausibility. Belief gives the most conservative probability among the consistent probability measures, so in our example $Bel(\{3\}) = 0.3$. In contrast, plausibility is the most optimistic probability among the consistent probability measures, hence in our example we have $Pl(\{3\}) = 0.5$. The observations in the example were represented by assigning masses to subsets of Ω : a mass of 0.5 to $\{1\}$, a mass of 0.3 to $\{3\}$, and a mass of 0.2 to $\{2, 3\}$. Such an assignment is called a basic belief assignment. Next we formalize these concepts.

2.4. Basic belief assignment, belief, and plausibility

The definitions in this section adopt the standard representations of belief functions. We use Meester and Slooten [4, Section 13.1] as a direct reference for this thesis, while the theory was originally developed by Shafer [6]. Let Ω be a finite outcome space, representing the set of possible outcomes in a probabilistic experiment. In classical probability theory, each individual outcome is assigned a probability. In Dempster-Shafer theory, however, a piece of evidence may support multiple outcomes without distinguishing between them. We therefore assign mass to subsets of Ω . Such an assignment is called a basic belief assignment and is defined on the set 2^Ω . The formal definition is then

Definition 2.3. A function $m : 2^\Omega \rightarrow [0, 1]$ is called a *basic belief assignment* if $m(\emptyset) = 0$ and

$$\sum_{C \subseteq \Omega} m(C) = 1.$$

Note here that if m is nonzero only on singletons (individual outcomes), then m corresponds to a regular probability measure. The basic belief assignment (bba) is the fundamental mass assignment from which belief and plausibility are constructed. For a given bba m , the subsets of Ω with positive mass are called its *focal elements*. As expressed before, the belief in an event is the most conservative assessment that does not conflict with the evidence at hand. To evaluate the belief in an event X , we only count the mass assigned to focal elements that guarantee that X has occurred.

Return to Example 2.1, where $m(\{1\}) = 0.5$, $m(\{3\}) = 0.3$, and $m(\{2, 3\}) = 0.2$. The ambiguous mass $m(\{2, 3\})$ does not contribute to $Bel(\{2\})$ or $Bel(\{3\})$, since it does not guarantee either individual outcome. It does contribute to $Bel(\{2, 3\})$, since the entire focal element $\{2, 3\}$ is contained in this event. Hence, $Bel(\{2\}) = 0$, $Bel(\{3\}) = 0.3$, and $Bel(\{2, 3\}) = 0.5$.

More generally, the belief in X is the sum of the masses assigned to all focal elements contained within X . This leads to the formal definition:

Definition 2.4. The *belief function* $Bel : 2^\Omega \rightarrow [0, 1]$ corresponding to the bba m is given by

$$Bel(X) = \sum_{Y \subseteq X} m(Y).$$

Finally, the plausibility function Pl gives the most optimistic probability consistent with the evidence. The plausibility of an event X therefore includes the mass of every focal element that intersects X . In Example 2.1, the focal element $\{2, 3\}$ intersects both $\{2\}$ and $\{3\}$. Its mass therefore contributes to the plausibility of both outcomes, giving $Pl(\{2\}) = 0.2$ and $Pl(\{3\}) = 0.5$.

Definition 2.5. The *plausibility* of an event X is defined as

$$Pl(X) := \sum_{Y \cap X \neq \emptyset} m(Y) = 1 - Bel(X^c).$$

The bba is therefore the foundational mass assignment, where the sets receiving mass determine the level of ambiguity. If m is nonzero only on singletons, then it corresponds to a regular probability measure and the different representations coincide at those singletons. Namely, for every singleton $X = \{a\}$, where $a \in \Omega$, we have $m(X) = Bel(X) = Pl(X) = P(X)$. At the other extreme, complete ignorance is represented by $m(\Omega) = 1$. In that case, for every nonempty proper subset $Y \subset \Omega$, we have $Bel(Y) = 0$ and $Pl(Y) = 1$.

To complete Example 2.1, the values of the basic belief assignment, belief, and plau-

sibility for every subset of Ω are

X	$m(X)$	$Bel(X)$	$Pl(X)$
$\emptyset$	0	0	0
$\{1\}$	0.5	0.5	0.5
$\{2\}$	0	0	0.2
$\{3\}$	0.3	0.3	0.5
$\{1, 2\}$	0	0.5	0.7
$\{1, 3\}$	0	0.8	1
$\{2, 3\}$	0.2	0.5	0.5
Ω	0	1	1.

For example, the mass on $\{2, 3\}$ contributes to $Bel(\{2, 3\})$, since it is contained in that event. The same mass contributes to both $Pl(\{2\})$ and $Pl(\{3\})$, since $\{2, 3\}$ intersects both singleton events.

2.5. Equivalent representations of belief

There are multiple equivalent representations for a belief function. That is, the same belief function can be represented through its bba m , through Bel , or through its collection $\mathcal{P}$ of consistent probability measures. We have seen how these definitions are related, but we have not yet established that they all uniquely define one another. This means that when we have one of these characterizations, then we can recover exactly the two remaining. To be precise, a belief function admits an equivalent representation as the lower envelope of its consistent probability measures, and its bba can be recovered uniquely through Möbius inversion [4, Section 13.1]. In this section we explain these equivalences in detail.

When a belief function Bel is given, define $\mathcal{P}$ to be the collection of probability measures consistent with Bel , meaning that $P(X) \geq Bel(X)$ for every $X \subseteq \Omega$. In general, $\mathcal{P}$ contains many distinct probability measures. It can be shown that Bel corresponds to the *lower envelope* of $\mathcal{P}$;

$$Bel(X) = \min\{P(X) : P \in \mathcal{P}\}.$$

Similarly, plausibility corresponds to the *upper envelope* of $\mathcal{P}$:

$$Pl(X) = \max\{P(X) : P \in \mathcal{P}\}.$$

To see why this is true, we can view the minimization of $P(X)$ as a routing of probability mass. Recall that any compatible probability measure $P \in \mathcal{P}$ is constructed by taking the mass $m(Y)$ of each focal element Y and in some way distributing it among its singletons. Hence to minimize the probability assigned to the set X , we assign as much mass as possible outside of X . If a focal element Y is contained within X (so $Y \subseteq X$),

the mass cannot be moved outside of X . The sum of all the trapped mass is precisely $\sum_{Y \subseteq X} m(Y)$. For any focal element that is not entirely contained within X , there is at least one outcome that lies outside of X . We can then put all of the mass of that focal element on an outcome outside of X . Hence, minimizing $P(X)$ results in distributing the mass such that $P(X)$ equals the sum of the masses of all focal elements contained in X , which equals $Bel(X)$.

This establishes that we can completely recover the belief function Bel when the collection $\mathcal{P}$ is known. We have already seen how to define $\mathcal{P}$ from Bel , so this completes the one-to-one correspondence between a belief function and its collection of consistent probabilities.

Next we turn to the basic belief assignment m . There it turns out too that we can reconstruct the exact basic belief assignment m from Bel . Let us illustrate this by returning to Example 2.1. Suppose we want to isolate the mass assigned specifically to $\{2, 3\}$. The value $Bel(\{2, 3\})$ contains the mass of $\{2, 3\}$ itself, together with the masses of all its proper subsets. Hence,

$$m(\{2, 3\}) = Bel(\{2, 3\}) - Bel(\{2\}) - Bel(\{3\}) + Bel(\emptyset) = 0.5 - 0 - 0.3 + 0 = 0.2.$$

Notice here the alternating plus and minus signs. If we want to determine the mass assigned to the entire outcome space Ω , when $|\Omega| = 3$, isolating its specific mass would require us to subtract the beliefs of all two-element subsets, but in doing so we would subtract the single-element subsets twice. To correct this, we would have to add the single-element beliefs back in. This alternating pattern of subtracting and adding subsets to isolate a specific value is a well-known mathematical mechanism called the inclusion-exclusion principle. For sets, applying this inclusion-exclusion mechanism is known as the Möbius transform. Through the Möbius transform, we can define the basic belief assignment entirely in terms of the belief function:

Definition 2.6. Given a belief function $Bel : 2^\Omega \rightarrow [0, 1]$, its corresponding basic belief assignment $m : 2^\Omega \rightarrow [0, 1]$ is recovered via the Möbius transform:

$$m(X) = \sum_{Y \subseteq X} (-1)^{|X \setminus Y|} Bel(Y).$$

The Möbius transform uniquely recovers m . We have already seen how to uniquely construct Bel from m , together this implies that there is a one-to-one correspondence between a belief function and its bba. From now on, we can treat these three representations as equivalent, regularly switching between them as one can always be derived from the other.

2.6. Design choices in defining conditional belief

Following the relative-frequency perspective on conditional belief presented by Meester and Slooten [4, Section 13.2], we now have a first look at conditional belief. Under the frequentist perspective, one can think of conditional probabilities from the viewpoint of

relative frequencies. Recall that from this perspective, the conditional probability of an event X given an event E is the relative frequency of X occurring in the subsequence determined by the occurrence of E . As explained in Section 2.2, $P(X \mid E)$ is interpreted as the relative frequency of X within the subsequence of trials where E occurred. Applying the corresponding formula

$$P(X \mid E) = \lim_{n \rightarrow \infty} \frac{\sum_{i=1}^n 1_{X \cap E}(x_i)}{\sum_{i=1}^n 1_E(x_i)}$$

naturally recovers the formula for conditional probability $P(X \mid E) = P(X \cap E)/P(E)$. Unfortunately, this perspective does not by itself provide a definition for conditional belief; this we can illustrate by revisiting Example 2.1. For building intuition, let us represent the relative frequencies by supposing that the experiment was repeated ten times. Five times $\{1\}$ was reported, three times $\{3\}$ was reported, and two times came the ambiguous report supporting $\{2, 3\}$.

Suppose we now wish to condition our knowledge on the event $E = \{1, 2\}$. If we attempt to apply the frequentist definition, we must determine the relative frequency of $\{1\}$ within the subsequence where E occurred, which requires us to count exactly how many times E occurred. For the 5 observations of $\{1\}$, the event E certainly occurred. For the 3 observations of $\{3\}$, E certainly did not occur. But for the 2 ambiguous observations of $\{2, 3\}$, we do not know whether to count them. If the true underlying face was 2, then E occurred; if the true face was 3, then E did not occur. From the consistent probability measures P_α from Example 2.1, we note $\frac{5}{7} \leq P_\alpha(\{1\} \mid E) \leq 1$. The lower value results when both ambiguous reports are counted as occurrences of outcome 2, while the upper value results when neither is included in the subsequence determined by E . Different definitions of conditional belief differ precisely in how they treat such ambiguous observations. In the next section, we introduce FH conditioning, which is adopted throughout this thesis.

2.7. Fagin-Halpern conditioning

Suppose Bel is a belief function defined on the finite space Ω . Recall that a belief function $Bel : 2^\Omega \rightarrow [0, 1]$ is given by $Bel(X) = \sum_{Y \subseteq X} m(Y)$ for some corresponding basic belief assignment m . As previously established, every belief function has a unique corresponding basic belief assignment m . Moreover, recall that a belief function Bel is the lower envelope of the collection $\mathcal{P}$ of all probability measures consistent with Bel . That is, $Bel(X)$ is the minimum of $P(X)$ for all $P \in \mathcal{P}$. So for our given belief function $Bel(\cdot)$, for all $X \subseteq \Omega$,

$$Bel(X) = \min\{P(X) : P \in \mathcal{P}\}.$$

We are interested in conditioning this belief on some set E . While several conditioning rules exist, we follow Fagin and Halpern [2] and condition every probability measure consistent with Bel , after which we take the resulting lower envelope.

Definition 2.7. Suppose Bel is a belief function on a finite set Ω , let $\mathcal{P}$ be the collection of probability measures consistent with Bel , and let $E \subseteq \Omega$ satisfy $Bel(E) > 0$. Then, for every $X \subseteq \Omega$,

$$Bel(X \mid E) := \min\{P(X \mid E) : P \in \mathcal{P}\}.$$

We refer to this as the *FH lower-envelope definition*. This attribution to Fagin and Halpern itself must be explained. Namely, an expression equivalent to the FH closed form already appears in Dempster's early work [1], while Dempster's standard conditioning rule is the distinct combination-based rule introduced below. Fagin and Halpern however adopted the lower-envelope construction explicitly as their definition of conditional belief, derived its closed-form expression, and proved that the resulting function is again a belief function [2]. For this reason we refer to the operation as FH conditioning.

We now turn to deriving this closed-form expression that does not depend on $\mathcal{P}$. As in the definition above, suppose that $Bel(E) > 0$. The following mass-routing derivation of the closed-form expression follows the method from Meester and Slooten [4, Section 13.2].

Recall that $P(X \mid E) = P(X \cap E)/P(E)$, and note that $P(E) = P(X \cap E) + P(E \setminus X)$. Hence,

$$Bel(X \mid E) = \min \left\{ \frac{P(X \cap E)}{P(X \cap E) + P(E \setminus X)} : P \in \mathcal{P} \right\}.$$

The fraction is increasing in $P(X \cap E)$ and decreasing in $P(E \setminus X)$. Hence, to minimize it, we must minimize $P(X \cap E)$ while simultaneously maximizing $P(E \setminus X) = P(X^c \cap E)$. Note however that this minimization and maximization should be done over $\mathcal{P}$, meaning we find a single measure $P \in \mathcal{P}$ satisfying both objectives. We now show that a single $P \in \mathcal{P}$ always exists that attains both extremes. By definition, we have $\min\{P(X \cap E) : P \in \mathcal{P}\} = Bel(X \cap E)$ so this is the absolute minimal value that can be attained. Furthermore, $\max\{P(E \setminus X) : P \in \mathcal{P}\} = Pl(X^c \cap E)$. Next we claim that both these extremes are attained by one $P \in \mathcal{P}$. Namely, any probability measure P compatible with a belief function is formed by taking the basic belief assignment $m(Y)$ of each focal set Y and distributing it among the elements $x \in Y$. To minimize $P(X \cap E)$ while simultaneously maximizing $P(E \setminus X) = P(X^c \cap E)$, we intuitively move mass towards the set $E \setminus X$ while moving mass away from $X \cap E$. Because the sets $X \cap E$ and $E \setminus X$ are mutually exclusive, these two objectives do not conflict. More precisely, whenever a focal element Y intersects $E \setminus X$, we assign its mass to an outcome in $Y \cap (E \setminus X)$. If Y does not intersect $E \setminus X$ but is not contained in $X \cap E$, we assign its mass to an outcome in $Y \setminus (X \cap E)$. Only when $Y \subseteq X \cap E$ is its mass forced to remain in $X \cap E$. This single routing therefore gives $P(X \cap E) = Bel(X \cap E)$ and $P(E \setminus X) = Pl(E \setminus X)$. Substituting these extremes yields

$$Bel(X \mid E) = \frac{\min\{P(X \cap E) : P \in \mathcal{P}\}}{\min\{P(X \cap E) : P \in \mathcal{P}\} + \max\{P(E \setminus X) : P \in \mathcal{P}\}} = \frac{Bel(X \cap E)}{Bel(X \cap E) + Pl(X^c \cap E)}.$$

Let us confirm this expression through Example 2.1. The closed-form expression gives $Bel(\{1\} \mid \{1, 2\}) = 0.5/(0.5 + 0.2) = 5/7$, agreeing with the lower bound found in the previous section.

2.8. Dempster conditioning

There exist different definitions for conditional belief. A lower-envelope expression equivalent to the one derived above already appears in Dempster's early work [1]. Dempster also developed a more general rule for combining sources of evidence, from which a different conditioning rule is obtained. The general question of when two such sources of evidence should be combined falls outside the scope of this thesis. We introduce the rule only to explain Dempster conditioning, and to show that when viewing belief as a tool in our epistemic toolbox, there are multiple epistemic interpretations leading to different definitions.

Dempster's rule combines basic belief assignments by intersecting their focal elements and normalizing the nonconflicting mass [1]. Let us show this precisely. Suppose we have two belief functions Bel_1 and Bel_2 on Ω , with corresponding basic belief assignments m_1 and m_2 . These belief functions may represent two different sources of evidence concerning the same unknown outcome. Dempster's rule then compares every focal element of m_1 with every focal element of m_2 . The masses are multiplied, and their product is assigned to the intersection of the two focal elements. If such an intersection is empty, the corresponding product is treated as conflicting mass. This conflict is discarded, after which the remaining masses are normalized back to one.

Definition 2.8. Let m_1 and m_2 be basic belief assignments on the same finite outcome space Ω . First define the unnormalized mass assignment $m_\cap : 2^\Omega \rightarrow [0, 1]$ by

$$m_\cap(C) := \sum_{\substack{A, B \subseteq \Omega \\ A \cap B = C}} m_1(A)m_2(B), \quad C \subseteq \Omega.$$

The value $m_\cap(\emptyset)$ is then the total conflicting mass. Provided $m_\cap(\emptyset) < 1$, the *Dempster combination* $m_1 \oplus m_2$ is defined by $(m_1 \oplus m_2)(\emptyset) := 0$, and for every nonempty $C \subseteq \Omega$,

$$(m_1 \oplus m_2)(C) := \frac{m_\cap(C)}{1 - m_\cap(\emptyset)}.$$

As a small example constructed for this thesis, imagine two sources of forensic evidence concerning suspects $\{1, 2, 3\}$. Suppose then that a footprint somewhat implicates suspects 1 and 2, giving $m_1(\{1, 2\}) = 0.6$, $m_1(\{1, 2, 3\}) = 0.4$. Furthermore, a found fabric could belong to suspects 2 and 3, resulting in $m_2(\{2, 3\}) = 0.3$, $m_2(\{1, 2, 3\}) = 0.7$. The only suspect implicated by both pieces of evidence is suspect 2, so this pair contributes $m_1(\{1, 2\})m_2(\{2, 3\}) = 0.6 \times 0.3 = 0.18$ to $m_\cap(\{2\})$. Without mapping out the remaining masses, we then see a specific modelling choice in Dempster's rule. Namely, neither source assigned any mass specifically to suspect 2, but when combined there is a mass specifically on this suspect. This may fit an epistemic assumption where suspect 2 is indeed the only outcome compatible with both pieces of evidence. However, the resulting combination is now more specific than either of the sources separately. This introduced specificity can in some cases give paradoxical results when integrated into a definition for conditional belief.

Through Dempster's conditioning, incoming evidence is treated as a second belief function. So suppose we learn with certainty that the outcome lies in some event $E \subseteq \Omega$. This new information is represented by a basic belief assignment m_E defined by $m_E(E) := 1$, and $m_E(X) = 0$ for every $X \neq E$. Combining m with m_E intersects every original focal element A with E . The mass on A is therefore transferred to $A \cap E$. For combination of a general m with this m_E , the unnormalized mass assignment satisfies

$$m_{\cap}(C) = \sum_{\substack{A \subseteq \Omega \\ A \cap E = C}} m(A), \quad C \subseteq E,$$

since m_E assigns positive mass only to E . In particular, all mass on focal elements disjoint from E becomes conflicting mass, giving

$$m_{\cap}(\emptyset) = \sum_{A \subseteq E^c} m(A) = Bel(E^c).$$

Hence,

$$1 - m_{\cap}(\emptyset) = 1 - Bel(E^c) = Pl(E).$$

Provided $Pl(E) > 0$, Dempster's conditioned basic belief assignment is then defined, for every nonempty $C \subseteq E$, by

$$m_D(C \parallel E) := (m \oplus m_E)(C) = \frac{m_{\cap}(C)}{1 - m_{\cap}(\emptyset)} = \frac{\sum_{\substack{A \subseteq \Omega \\ A \cap E = C}} m(A)}{Pl(E)}. \quad (2.1)$$

This then leads to Dempster's definition of conditional belief.

Definition 2.9. Let Bel be a belief function on Ω and let $E \subseteq \Omega$ satisfy $Pl(E) > 0$. *Dempster conditional belief* is then defined by

$$Bel_D(X \parallel E) := \frac{Bel(X \cup E^c) - Bel(E^c)}{1 - Bel(E^c)}. \quad (2.2)$$

The key point for now is that a different epistemic viewpoint leads to a different definition. Dempster conditioning produces a belief function, while the following chapters will show that the same holds for FH conditioning. Moreover, when the original belief function is a regular probability measure, then both definitions reduce to the definition of conditional probability $P(X \mid E) = P(X \cap E)/P(E)$. Where the definitions differ is how they treat a focal element that partially overlaps E . Dempster conditioning bluntly transfers the entire mass of such a focal element onto its intersection with E . That is, if $m(A) > 0$ and both $A \cap E$ and $A \cap E^c$ are nonempty, then the complete mass $m(A)$ is treated as support for $A \cap E$ after conditioning. FH conditioning is more cautious. It takes into account that the mass on A could have been distributed in many different ways among the elements of A before conditioning. Some consistent probability measures may have placed almost all mass outside of E , while Dempster conditioning does not retain

these different distributions when transferring the full mass to $A \cap E$. Fagin and Halpern show that this distinction has a general implication. Namely, their paper shows that Dempster conditioning resolves more ambiguity than FH conditioning [2]. That is, if we let $Pl_D(\cdot \parallel E)$ and $Pl(\cdot \mid E)$ denote the plausibility functions corresponding to Dempster and FH conditional belief, respectively, then for every event X we have

$$Bel(X \mid E) \leq Bel_D(X \parallel E) \leq Pl_D(X \parallel E) \leq Pl(X \mid E).$$

So Dempster conditioning is less conservative, producing a higher lower bound and a lower upper bound for the conditional probability of every event. This does not necessarily mean that Dempster conditioning is more accurate. The narrower interval is obtained by resolving part of the original ambiguity through intersection. This may be justified by the meaning of the new information, but it can also create unwarranted confidence.

The three prisoners problem

Fagin and Halpern illustrate the difference between the two conditioning rules through the paradoxical three prisoners problem [2, pp. 359–360]. Suppose that two of three prisoners a, b, c will be executed, each with equal probability, but a does not know which. He then says to the jailer, “Since either b or c is certainly going to be executed, you will give me no information about my own chances if you tell me the name of one of the other prisoners to be executed”. Once the jailer answers that b will be executed, prisoner a is relieved, believing that his chance of survival has improved from $1/3$ to $1/2$.

Whether or not a is correct in this assessment depends completely on how the jailer decides who to name. That is, if c was to be pardoned, then the jailer had no choice but to say b . If however a is pardoned, then both b and c will be executed so the jailer may name either one. The problem does not specify how the jailer makes this choice. Let us now test our definitions on this problem.

Following Fagin and Halpern, let (x, y) denote that prisoner x is pardoned and the jailer names prisoner y . Hence define the focal sets

$$L_a = \{(a, b), (a, c)\}, \quad L_b = \{(b, c)\}, \quad L_c = \{(c, b)\},$$

and assign mass $1/3$ to each. Note that the mass on L_a is not divided between (a, b) and (a, c) , because the jailer’s behaviour when a is pardoned is unknown.

Let now $S_b = \{(a, b), (c, b)\}$ be the event that the jailer says b . Dempster conditioning intersects every focal element with S_b . Since $L_b \cap S_b = \emptyset$, the mass on L_b becomes conflicting mass. After discarding this mass, the masses on L_a and L_c are each concentrated on one remaining outcome. After normalization this gives

$$Bel_D(L_a \parallel S_b) = Pl_D(L_a \parallel S_b) = \frac{1}{2}.$$

Thus, Dempster conditioning reduces the interval to the single value $1/2$. FH conditioning instead gives

$$Bel(L_a \mid S_b) = 0, \quad Pl(L_a \mid S_b) = \frac{1}{2}.$$

To see the meaning of this interval, we again follow Fagin and Halpern and let α be the probability that the jailer says b when a is pardoned. The conditional probability that a is pardoned then equals $\alpha/(1 + \alpha)$, which ranges from 0 to $1/2$ as α ranges from 0 to 1.

The Dempster value $1/2$ therefore corresponds to the specific case $\alpha = 1$, where the jailer says b when a is pardoned. This value is not necessarily wrong, but it builds on an assumption that was not explicitly present in the problem. Dempster conditioning introduces it by transferring all mass on $\{(a, b), (a, c)\}$ onto its intersection $\{(a, b)\}$. FH conditioning is more cautious because it accounts for all possible behaviors consistent with the original information.

Dempster conditioning resolves ambiguity by intersecting focal elements with the new evidence, whereas FH conditioning retains all conditional probabilities compatible with the original information. Neither of the approaches is therefore universally correct, as both approaches contain implicit assumptions about how to deal with ambiguous support. This is precisely illustrated in the example, as Dempster conditioning corresponds to a particular jailer protocol, while FH conditioning accounts for the full range of protocols compatible with the problem statement.

That concludes our discussion on Dempster conditioning. For the remainder of this thesis, conditional belief will refer to FH conditional belief. Accordingly, throughout the remainder of this thesis, $Bel(\cdot \mid E)$ denotes the function obtained through FH conditioning. In particular, for every $X \subseteq \Omega$,

$$Bel(X \mid E) = \frac{Bel(X \cap E)}{Bel(X \cap E) + Pl(X^c \cap E)}.$$

We refer to this formula as the *closed-form expression for FH conditional belief*, or briefly as the *FH closed form*.

2.9. Central theorem

This completes our discussion on the foundations of conditional belief. Hence, we are now ready to present the central theorem of this thesis.

Theorem 2.10. *Let Bel be a belief function defined on a finite outcome space Ω . Let $E \subseteq \Omega$ be an event such that $Bel(E) > 0$. Then $Bel(\cdot \mid E)$ is a belief function.*

The remainder of the thesis investigates two proofs of this theorem, one by Fagin and Halpern [2], and another by Jaffray [3]. Both proofs entail constructing a conditional basic belief assignment for the function $Bel(\cdot \mid E)$. We will be analyzing these proofs, focusing specifically on how these constructions work mathematically, and how the constructions can be interpreted epistemically.

3. Fagin and Halpern

In this chapter we will present the proof by Fagin and Halpern [2] showing that conditional belief is itself a belief function. We follow their proof structure and auxiliary mass construction, while the notation is adapted to have consistent notation throughout this thesis. The examples in Section 3.2 are constructed for this thesis to explain how the FH definitions may have been created.

3.1. Proof outline and defining focal sets

We can sufficiently prove that a function is a belief function by finding its corresponding basic belief assignment. Following Fagin and Halpern [2], we first restrict the conditional belief to subsets of E . Let $Bel' : 2^E \rightarrow [0, 1]$ be defined by $Bel'(X) := Bel(X \mid E)$ for $X \subseteq E$, and let $m' : 2^E \rightarrow \mathbb{R}$ be its candidate mass function. To prove Theorem 2.10, we must verify the following four conditions:

- A. $m'(\emptyset) = 0$,
- B. $m'(X) \geq 0 \quad \forall X \subseteq E$,
- C. $\sum_{X \subseteq E} m'(X) = 1$,
- D. $Bel'(X) = \sum_{Y \subseteq X} m'(Y) \quad \forall X \subseteq E$.

Conditions B and C together also imply that $m'(X) \leq 1$ for every $X \subseteq E$. Once these conditions are established, the result on 2^Ω follows from $Bel(X \mid E) = Bel'(X \cap E)$.

Before diving into the details of the proof it will be useful to first build an intuition for how the basic belief assignment for conditional belief should behave. Then we want to condition on the set of outcomes E . For any set X that does not intersect E , it is easily verified that the conditional belief given E must be 0; that is, $X \subseteq \Omega \setminus E \implies Bel(X \mid E) = 0$. Conversely, for sets containing E , the conditional belief given E is 1; that is, $E \subseteq X \implies Bel(X \mid E) = 1$. For sets contained in E , their conditional belief relies on the plausibility of the remaining space in $E \setminus X$. Specifically, if $X \subseteq E$, applying the FH closed form yields $Bel'(X) = \frac{Bel(X)}{Bel(X) + Pl(E \setminus X)}$. These three cases should be intuitively interpretable analogously to conditional probability. The remaining sets partially overlap E , their conditional belief is harder to assess and they will therefore be the main sets of interest in this proof, and so we will create a labeling of sets based on this distinction.

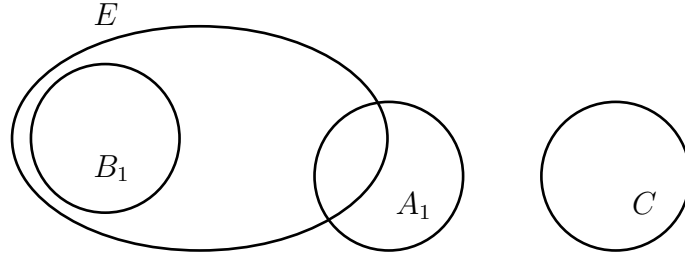
Let $\mathcal{B} := \{B_1, \dots, B_t\}$ be the collection of focal elements that are contained within E . Note that we have at least one such B_i , as we assumed earlier that $Bel(E) > 0$.

Furthermore, let $\mathcal{A} := \{A_1, \dots, A_n\}$ be the collection of focal elements that overlap both E and its complement $\Omega \setminus E$. For these overlapping focal elements, let $A'_i = A_i \cap E$ denote their respective intersections with the conditioning event, for $1 \leq i \leq n$.

3.2. Reverse-engineering bba through examples

The examples in this section are not taken from Fagin and Halpern, they are constructed here to reverse-engineer their auxiliary function m'' . We first illustrate why this categorization of sets is helpful through an example.

Example 3.1. So consider the setting where the bba on Ω has three focal elements, and we wish to condition on an event E . Let B_1 be a focal element entirely contained within E , let A_1 be a focal element that partially overlaps E , and let C be a focal element entirely disjoint from E . Suppose their respective masses are $m(B_1)$, $m(A_1)$, and $m(C)$, such that $m(B_1) + m(A_1) + m(C) = 1$. We are then interested in the values $Bel(\cdot | E)$ takes, as well as the basic belief assignment m' that gives rise to it. A Venn diagram is shown below.



Consider now the conditional beliefs involved. First, let us look at the belief in the set C . Applying the FH lower-envelope definition gives

$$Bel(C | E) = \min \left\{ \frac{P(C \cap E)}{P(E)} : P \in \mathcal{P} \right\},$$

where $C \cap E = \emptyset$ and so $P(C \cap E) = 0$, which implies $Bel(C | E) = 0$. Consequently, when the resulting bba is extended from 2^E to 2^Ω , every subset of C must receive mass 0.

Consider next any subset $X \subseteq E$ that does not contain B_1 . In our example, neither X nor any of its subsets can contain a focal element, yielding $Bel(X) = 0$, from which it follows that $Bel(X | E) = 0$.

Next consider

$$Bel(B_1 | E) = \min \left\{ \frac{P(B_1 \cap E)}{P(E)} : P \in \mathcal{P} \right\} = \min \left\{ \frac{P(B_1)}{P(B_1) + P(E \setminus B_1)} : P \in \mathcal{P} \right\}.$$

As we saw before, this means the minimal P over $\mathcal{P}$ will have minimized the mass distributed on the set B_1 , and maximized the mass on the set $E \setminus B_1$. Hence we see

$Bel(B_1 | E) = \frac{m(B_1)}{m(B_1)+m(A_1)}$. Now from before we knew that all proper subsets of B_1 have zero basic belief assignment, so we can conclude that $m'(B_1) = \frac{m(B_1)}{m(B_1)+m(A_1)}$.

Next, let us check the set A'_1 .

$$Bel(A'_1 | E) = \min \left\{ \frac{P(A'_1 \cap E)}{P(E)} : P \in \mathcal{P} \right\} = \min \left\{ \frac{P(A'_1)}{P(A'_1) + P(E \setminus A'_1)} : P \in \mathcal{P} \right\}.$$

Now $P(A'_1)$ is minimized by putting all mass of the focal element A_1 onto the part $A_1 \setminus E$, resulting in $P(A'_1) = 0$. Hence $Bel(A'_1 | E) = 0$, so the basic belief assignment m' of the set A'_1 and all its subsets must equal 0.

Consider now the set $B_1 \cup A'_1$; we then see

$$\begin{aligned} Bel(B_1 \cup A'_1 | E) &= \min \left\{ \frac{P((B_1 \cup A'_1) \cap E)}{P(E)} : P \in \mathcal{P} \right\} \\ &= \min \left\{ \frac{P(B_1 \cup A'_1)}{P(B_1 \cup A'_1) + P(E \setminus (B_1 \cup A'_1))} : P \in \mathcal{P} \right\}. \end{aligned}$$

From m , we know $P(B_1 \cup A'_1)$ is minimized at $m(B_1)$, and $P(E \setminus (B_1 \cup A'_1)) = 0$ for all $P \in \mathcal{P}$. Hence we see $Bel(B_1 \cup A'_1 | E) = \frac{m(B_1)}{m(B_1)} = 1$. Since $Bel(B_1 \cup A'_1 | E) = m'(B_1) + m'(A'_1) + m'(B_1 \cup A'_1)$, and we previously established $m'(A'_1) = 0$, we can conclude that $m'(B_1 \cup A'_1)$ must equal $\frac{m(A_1)}{m(B_1)+m(A_1)}$.

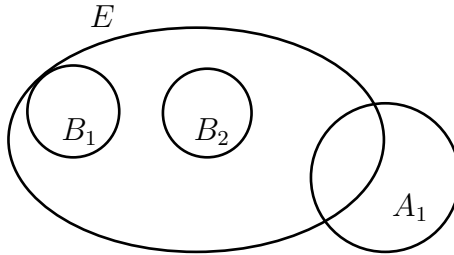
What this example indicates is that in considering the function $Bel(\cdot | E)$, we need not consider the masses of sets outside of E . Hence, to create m' we can exclude those masses and renormalize; this will prove useful later. Let $N = \sum_{i=1}^n m(A_i) + \sum_{i=1}^t m(B_i)$. Note that $N > 0$ since there is at least one B_i . We then define the normalized mass variables as $a_i = \frac{m(A_i)}{N}$ and $b_i = \frac{m(B_i)}{N}$ for each i .

Furthermore, note that we can restrict the domain of $Bel(\cdot | E)$ to just the subsets of E to simplify notations. Namely, for any arbitrary event $X \subseteq \Omega$, we have that

$$\begin{aligned} Bel(X | E) &= \frac{Bel(X \cap E)}{Bel(X \cap E) + Pl(X^c \cap E)} \\ &= \frac{Bel(X \cap E)}{Bel(X \cap E) + Pl((X \cap E)^c \cap E)} = Bel(X \cap E | E). \end{aligned}$$

We continue showing what values $Bel(\cdot | E)$ and its updated basic belief assignment m' should be with another example.

Example 3.2. Consider the setting in the Venn diagram below.



So we now have three sets of interest: the sets B_1 , B_2 , and A_1 . We then see

$$Bel(B_1 | E) = \min \left\{ \frac{P(B_1)}{P(B_1) + P(E \setminus B_1)} : P \in \mathcal{P} \right\} = \frac{m(B_1)}{m(B_1) + m(B_2) + m(A_1)}.$$

Using our normalized variables, this fraction simplifies to $\frac{b_1}{b_1 + b_2 + a_1} = b_1$. As in the previous example, any proper subset X of B_1 by assumption has $Bel(X | E) = 0$; from this we conclude that we must have $m'(B_1) = b_1$. Symmetrically, we have $m'(B_2) = b_2$.

Next consider the set A'_1 . We have

$$Bel(A'_1 | E) = \min \left\{ \frac{P(A'_1)}{P(A'_1) + P(E \setminus A'_1)} : P \in \mathcal{P} \right\}.$$

Now $P(A'_1)$ is minimized by putting all mass of the focal element A_1 outside of E onto the part $A_1 \setminus E$, resulting in $P(A'_1) = 0$. Hence $Bel(A'_1 | E) = 0$, so the basic belief assignment of the set A'_1 and all its subsets must equal 0.

We continue with the set $B_1 \cup A'_1$. There we see

$$Bel(B_1 \cup A'_1 | E) = \min \left\{ \frac{P(B_1 \cup A'_1)}{P(B_1 \cup A'_1) + P(E \setminus (B_1 \cup A'_1))} : P \in \mathcal{P} \right\}.$$

From m , we know $P(B_1 \cup A'_1)$ is minimized at $m(B_1)$, and $P(E \setminus (B_1 \cup A'_1))$ attains a maximal value of $m(B_2)$. Hence we see $Bel(B_1 \cup A'_1 | E) = \frac{m(B_1)}{m(B_1) + m(B_2)} = \frac{b_1}{b_1 + b_2} = \frac{b_1}{1 - a_1}$. From there, since $Bel(B_1 \cup A'_1 | E) = m'(B_1) + m'(A'_1) + m'(B_1 \cup A'_1)$, we can conclude that $m'(B_1 \cup A'_1)$ must equal $\frac{b_1}{1 - a_1} - b_1$. By symmetry, we find $m'(B_2 \cup A'_1) = \frac{b_2}{1 - a_1} - b_2$.

So far we have assigned masses to the sets B_1 , B_2 , $B_1 \cup A'_1$, and $B_2 \cup A'_1$. Let us now sum the masses assigned so far:

$$b_1 + b_2 + \frac{b_1}{1 - a_1} - b_1 + \frac{b_2}{1 - a_1} - b_2 = \frac{b_1 + b_2}{1 - a_1} = 1.$$

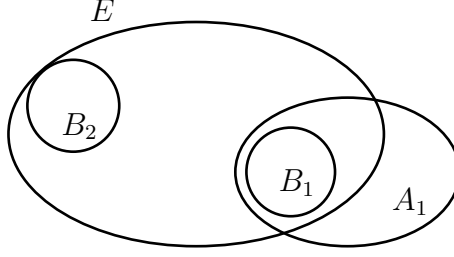
So, if m' is a valid belief assignment, then no other set can have a positive mass. Let us investigate this by considering $m'(B_1 \cup B_2)$. We have

$$\begin{aligned} Bel(B_1 \cup B_2 | E) &= \min \left\{ \frac{P(B_1 \cup B_2)}{P(B_1 \cup B_2) + P(E \setminus (B_1 \cup B_2))} : P \in \mathcal{P} \right\} \\ &= \frac{m(B_1) + m(B_2)}{m(B_1) + m(B_2) + m(A_1)} = b_1 + b_2 = m'(B_1) + m'(B_2). \end{aligned}$$

Hence we see that indeed $m'(B_1 \cup B_2) = 0$. Similarly, it is easy to check that also $m'(B_1 \cup B_2 \cup A'_1) = 0$. So indeed, in this example we find that we can use FH conditional belief to construct some m' which so far has been nonnegative, with masses of subsets of E summing to 1.

Something peculiar occurs when we have the situation $B_1 = A'_1$. We look into this in the next example.

Example 3.3. Consider the setting shown in the Venn diagram below.



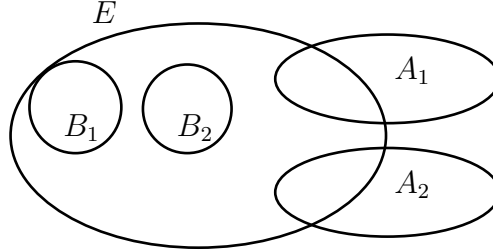
As before, one might expect that $m'(B_1) = b_1$. However, this turns out not to be the case. Namely, consider

$$\text{Bel}(B_1 | E) = \min \left\{ \frac{P(B_1)}{P(B_1) + P(E \setminus B_1)} : P \in \mathcal{P} \right\} = \frac{b_1}{b_1 + b_2} = \frac{b_1}{1 - a_1}.$$

Here, $P(E \setminus B_1)$ attains a maximal value of $m(B_2)$, whereas before this would have been $m(B_2) + m(A_1)$. Because we now have $B_1 = A'_1$, the probability measure P cannot shift mass from the focal element A_1 to the inside of E without changing the mass on the set B_1 . This poses a challenge in defining m' , which we will return to later.

The examples serve to understand how conditional belief behaves, and to identify the basic belief assignment that gives rise to it. We will explicitly define this basic belief assignment later, but first we consider one final example to see whether a general pattern emerges.

Example 3.4. We now introduce one more focal element partially inside of E ; the resulting Venn diagram is shown below.



Here we do not have the boundary case where $B_1 = A'_1$. Through the same methods as before, we can verify that $\text{Bel}(B_1 | E) = \frac{m(B_1)}{m(B_1) + m(B_2) + m(A_1) + m(A_2)} = b_1$, which implies $m'(B_1) = b_1$. Similarly to before, we find that $\text{Bel}(B_1 \cup A'_1 | E) = \frac{b_1}{1 - a_1}$; from this it follows that $m'(B_1 \cup A'_1) = \frac{b_1}{1 - a_1} - b_1$. Note that $a_1 > 0$ implies $m'(B_1 \cup A'_1) > 0$. By symmetry, we also have $m'(B_1 \cup A'_2) = \frac{b_1}{1 - a_2} - b_1$. Now we are interested in $m'(B_1 \cup A'_1 \cup A'_2)$, which by definition equals $\text{Bel}(B_1 \cup A'_1 \cup A'_2 | E) - m'(B_1 \cup A'_1) - m'(B_1 \cup A'_2) - m'(B_1)$. We first consider

$$\text{Bel}(B_1 \cup A'_1 \cup A'_2 | E) = \min \left\{ \frac{P(B_1 \cup A'_1 \cup A'_2)}{P(B_1 \cup A'_1 \cup A'_2) + P(E \setminus (B_1 \cup A'_1 \cup A'_2))} : P \in \mathcal{P} \right\}$$

$$= \frac{m(B_1)}{m(B_1) + m(B_2)} = \frac{b_1}{b_1 + b_2} = \frac{b_1}{1 - a_1 - a_2}.$$

Hence, we find

$$\begin{aligned} m'(B_1 \cup A'_1 \cup A'_2) &= \frac{b_1}{1 - a_1 - a_2} - \left(\frac{b_1}{1 - a_1} - b_1 \right) - \left(\frac{b_1}{1 - a_2} - b_1 \right) - b_1 \\ &= \frac{b_1}{1 - a_1 - a_2} - \frac{b_1}{1 - a_1} - \frac{b_1}{1 - a_2} + b_1. \end{aligned}$$

By now it should be clear that an inclusion-exclusion mechanism is at work; this is expected from the fundamental definition related to the Möbius transform $Bel(X) = \sum_{Y \subseteq X} m(Y)$. Also we see a somewhat repeating pattern in the fractions. Now the question arises: is this calculated value for $m'(B_1 \cup A'_1 \cup A'_2)$ nonnegative? In this specific case, we can see that it is, through the chain

$$\begin{aligned} \frac{b_1}{1 - a_1 - a_2} - b_1 &= \frac{b_1(a_1 + a_2)}{1 - a_1 - a_2} = \frac{b_1 a_1}{1 - a_1 - a_2} + \frac{b_1 a_2}{1 - a_1 - a_2} \\ &> \frac{b_1 a_1}{1 - a_1} + \frac{b_1 a_2}{1 - a_2} = \left(\frac{b_1}{1 - a_1} - b_1 \right) + \left(\frac{b_1}{1 - a_2} - b_1 \right). \end{aligned}$$

Thus we see that nonnegativity holds in this simple example. Proving that this holds generally will be the largest hurdle in the formal proof below.

3.3. Partial proof by Fagin and Halpern

We now return to the proof of Fagin and Halpern [2]. We first present the full statement that they prove, which deviates slightly from our central theorem. Suppose Bel is a belief function on a finite set Ω , and let $E \subseteq \Omega$ be an event such that $Bel(E) > 0$. For each $X \subseteq \Omega$, we express FH conditional belief through the FH closed form:

$$Bel(X | E) := \frac{Bel(X \cap E)}{Bel(X \cap E) + Pl(X^c \cap E)}.$$

Then $Bel(\cdot | E)$ is a belief function.

To prove this, they first define the function Bel' on the domain 2^E , using the FH closed form, so that for each subset X of E we have

$$Bel'(X) := \frac{Bel(X)}{Bel(X) + Pl(X^c \cap E)}.$$

They then define a candidate basic belief assignment m' and show it satisfies four conditions for all $X \subseteq E$:

A. $m'(\emptyset) = 0,$

- B. $m'(X) \geq 0$,
- C. $\sum_{X \subseteq E} m'(X) = 1$,
- D. $Bel'(X) = \sum_{Y \subseteq X} m'(Y)$.

The idea here is that subsets of $\Omega \setminus E$ will always have zero conditional belief, so the function Bel' only needs to be defined on E and its subsets. For any set $X \subseteq \Omega$, we then have $Bel(X | E) = Bel'(X \cap E)$; hence, to prove that $Bel(\cdot | E)$ is a belief function, it suffices to show this for Bel' .

We now adopt the same naming and normalization conventions established in the previous examples. Recall that $\mathcal{B} = \{B_1, \dots, B_t\}$ contains the focal elements enclosed within E , and $\mathcal{A} = \{A_1, \dots, A_n\}$ contains the focal elements partially overlapping E , with their intersections denoted as $A'_i = A_i \cap E$. Furthermore, let $N = \sum_{i=1}^n m(A_i) + \sum_{i=1}^t m(B_i)$ be our normalization constant. We then denote their respective normalized mass variables as $a_i = \frac{m(A_i)}{N}$ and $b_i = \frac{m(B_i)}{N}$.

It will now turn out that the only sets that can receive positive mass under m' are sets of the form $B_i \cup A'_{j_1} \cup \dots \cup A'_{j_s}$; this again is in line with what we have seen in the examples. However, as in the Example 3.3, something peculiar occurs when two sets of that form describe the same set. For instance, this happens when a subset X equals both $B_1 \cup A'_1$ and $B_2 \cup A'_3 \cup A'_4$. Additional notation is needed to handle such cases; we simplify the representation $B_i \cup A'_{j_1} \cup \dots \cup A'_{j_s}$ to $B_i A'_{j_1} \dots A'_{j_s}$, which we then call a *string*.

So suppose we have a string $s_1 \dots s_k$, and if $1 \leq i_1 < \dots < i_p \leq k$, then we call $s_{i_1} \dots s_{i_p}$ a substring of $s_1 \dots s_k$, for this we write $s_{i_1} \dots s_{i_p} \preccurlyeq s_1 \dots s_k$ ($\prec$ if proper). What this notation allows us to do is to denote all proper subsets of positive basic belief assignment for a set like $B_i \cup A'_{j_1} \cup \dots \cup A'_{j_s}$. Namely, that would be $\{B_i Y : Y \prec A'_{j_1} \dots A'_{j_s}\}$.

With that notation established, we can define a function m'' on strings of the form $B_i A'_{j_1} \dots A'_{j_s}$, where the A'_j terms do not repeat. We then define $m'(X) := \sum_{s \text{ represents } X} m''(s)$. Here ‘represents’ indicates that for a given set X , we take all possible strings $s = B_i A'_{j_1} \dots A'_{j_s}$ such that $X = B_i \cup A'_{j_1} \cup \dots \cup A'_{j_s}$, and sum their individual m'' values. Note that if no such duplicates in representation exist, then $m'(X)$ simply equals the m'' value of its unique representing string. Hence, in all our examples besides Example 3.3, the values of m' and m'' are identical.

Now we can explicitly define m'' . First, we let $m''(B_i) = b_i$ for all $1 \leq i \leq t$. We then require $m''(B_i A'_{j_1} \dots A'_{j_s})$ to be nonzero only when $j_1 < \dots < j_s$. This ensures that each fixed choice of B_i and indices $j_1, \dots, j_s$ has only one nonzero ordering. Different choices may still represent the same set, which is why $m'(X)$ sums over all strings representing X . Furthermore, since $Bel(E) > 0$, at least one b_i is positive. Since the normalized masses sum to 1, this implies that $\sum_{j=1}^n a_j < 1$. Hence, every denominator in the following definition is positive. Let then

$$m''(B_i A'_{j_1} \dots A'_{j_r}) = \frac{b_i}{1 - a_{j_1} - \dots - a_{j_r}} - \sum_{Y \prec A'_{j_1} \dots A'_{j_r}} m''(B_i Y). \quad (3.1)$$

In this fraction we recognize the inclusion exclusion mechanism that we saw in the examples, and also the shape of the fraction we have seen before.

We are now ready to verify conditions A through D, starting with condition A ($m'(\emptyset) = 0$). Note that the empty set is not represented by any set of the form $B_i \cup A'_{j_1} \cup \dots \cup A'_{j_s}$ because $B_i \neq \emptyset$ for all i . Hence, by our definition, $m'(\emptyset)$ is just an empty sum which evaluates to 0 as required.

Next we prove condition D ($Bel'(X) = \sum_{Y \subseteq X} m'(Y)$). Let us take $A'_{k_1}, \dots, A'_{k_q}$ to be the intersections A'_i contained in X , and let $B_{i_1}, \dots, B_{i_s}$ be the B_i 's contained in X . Recall that $Bel(X) = \sum_{Y \subseteq X} m(Y)$. Since $X \subseteq E$, no focal element A_i or focal element disjoint from E is contained in X . Hence we see that $Bel(X) = m(B_{i_1}) + \dots + m(B_{i_s})$. Now we are also interested in $Pl(X^c \cap E)$. As before, let $N = \sum_{i=1}^n m(A_i) + \sum_{i=1}^t m(B_i)$, so that N is the total mass relevant to E . Then the plausibility of $X^c \cap E$ is this total mass N minus the mass of the focal elements whose intersection with E falls fully inside X . Since $A'_{k_1}, \dots, A'_{k_q}, B_{i_1}, \dots, B_{i_s}$ are exactly these subsets of $X \subseteq E$, we find that

$$Pl(X^c \cap E) = N - (m(A_{k_1}) + \dots + m(A_{k_q}) + m(B_{i_1}) + \dots + m(B_{i_s})).$$

Now we can evaluate

$$Bel'(X) = \frac{Bel(X)}{Bel(X) + Pl(X^c \cap E)} = \frac{m(B_{i_1}) + \dots + m(B_{i_s})}{N - (m(A_{k_1}) + \dots + m(A_{k_q}))}.$$

Dividing both the numerator and denominator by N we then see that

$$Bel'(X) = \frac{Bel(X)}{Bel(X) + Pl(X^c \cap E)} = \frac{b_{i_1} + \dots + b_{i_s}}{1 - (a_{k_1} + \dots + a_{k_q})}.$$

To prove D, we must then show the equivalence

$$\sum_{Y \subseteq X} m'(Y) = \frac{b_{i_1} + \dots + b_{i_s}}{1 - (a_{k_1} + \dots + a_{k_q})}.$$

Let us call the expression $m''(B_i A'_{j_1} \dots A'_{j_r})$ a good term when i is a member of $\{i_1, \dots, i_s\}$, and where $j_1 < \dots < j_r$ are members of $\{k_1, \dots, k_q\}$. Note then that if $m''(B_i A'_{j_1} \dots A'_{j_r})$ is a good term, then $B_i \cup A'_{j_1} \cup \dots \cup A'_{j_r} \subseteq X$. Hence, $\sum_{Y \subseteq X} m'(Y)$ is the sum of all good terms. Namely, each good term is a part of the sum defining $m'(Y)$ for exactly one $Y \subseteq X$, and if $Y \subseteq X$ then $m'(Y)$ is defined as the sum of certain good terms. If $q = 0$, then the only good term beginning with B_i is $m''(B_i) = b_i$, so the desired identity follows directly. We may therefore assume that $q \geq 1$. Let now i be a fixed member of $\{i_1, \dots, i_s\}$. The sum of all good terms of the form $m''(B_i A'_{j_1} \dots A'_{j_r})$ except for the good term $m''(B_i A'_{k_1} \dots A'_{k_q})$ is simply $\sum_{Y \prec A'_{k_1} \dots A'_{k_q}} m''(B_i Y)$. Since

$$m''(B_i A'_{j_1} \dots A'_{j_r}) = \frac{b_i}{1 - a_{j_1} - \dots - a_{j_r}} - \sum_{Y \prec A'_{j_1} \dots A'_{j_r}} m''(B_i Y),$$

it then follows that the sum of all good terms of the form $m''(B_i A'_{j_1} \dots A'_{j_r})$ equals $b_i/(1 - a_{k_1} - \dots - a_{k_q})$. Summing this over all $i \in \{i_1, \dots, i_s\}$, we obtain

$$\sum_{Y \subseteq X} m'(Y) = \sum_{\ell=1}^s \frac{b_{i_\ell}}{1 - a_{k_1} - \dots - a_{k_q}} = \frac{b_{i_1} + \dots + b_{i_s}}{1 - a_{k_1} - \dots - a_{k_q}} = Bel'(X).$$

This proves D.

We continue with condition C ($\sum_{X \subseteq E} m'(X) = 1$). First we show that $Bel'(E) = 1$. By definition we have

$$Bel'(E) = \frac{Bel(E)}{Bel(E) + Pl(E^c \cap E)} = \frac{Bel(E)}{Bel(E)} = 1.$$

Then recall from D that we have shown $Bel'(X) = \sum_{Y \subseteq X} m'(Y)$, hence it follows that $\sum_{X \subseteq E} m'(X) = 1$.

3.4. Limitations of algebraic approach

What then remains is to prove the hardest requirement B ($m'(X) \geq 0 \forall X \subseteq E$). Recall that m' is defined as a sum of m'' terms. This recursive definition of m'' features a single positive leading term followed by a series of subtracted recursive calls. To show that this recursive formulation yields a non-negative value, Fagin and Halpern [2] derive an explicit formula for m'' . For a general string $B_1 A'_1 \dots A'_r$ they show this explicit formula to be

$$m''(B_1 A'_1 \dots A'_r) = \sum_{I \subseteq \{1, \dots, r\}} (-1)^{r-|I|} \frac{b_1}{1 - \sum_{j \in I} a_j}.$$

From this explicit formulation we immediately recognize the inclusion-exclusion pattern. For the purposes of this thesis, the alternating signs make the nonnegativity of m'' difficult to see directly. FH prove nonnegativity through a geometric series expansion and a double induction using derivatives. This establishes the result, but does not clarify how the positive conditional mass is generated. The overarching aim of this thesis is to explore not if, but why the central theorem holds. Since the technical argument used by FH does not help in our conceptual understanding, the remainder of the proof of requirement B has been moved to Appendix A. Jaffray's proof in the next chapter introduces an alternative construction in which the nonnegativity is structurally enforced.

4. Jaffray

In Chapter 3, the bba m' corresponding to FH conditional belief was reconstructed through the auxiliary function m'' . The inclusion-exclusion structure made nonnegativity of the conditioned bba difficult to prove. In this chapter, we reconstruct Jaffray's alternative proof [3]. Jaffray also creates an auxiliary function to construct the conditioned bba m' , but his function is defined on ordered sequences instead of on strings. The proof structure and ordered-sequence construction are due to Jaffray, while the examples in Section 4.1 and Section 4.4 are developed for this thesis to illustrate the construction. We first illustrate this change in construction with an example, then we present Jaffray's proof, and finally work through a large example showcasing the entire proof process.

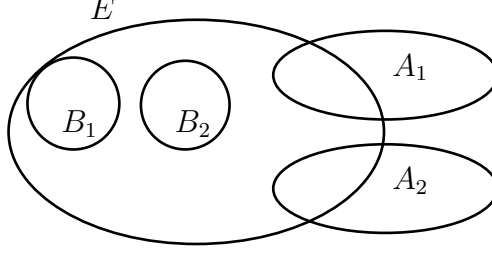
4.1. From strings to ordered sequences

Both FH and Jaffray reconstruct the basic belief assignment m' corresponding to $Bel(\cdot | E)$ by identifying a recursive structure in their respective auxiliary functions. Their difference lies mainly in the bookkeeping. The recursion by FH is based on inclusion-exclusion over strings, whereas Jaffray retains the order in which partially overlapping focal elements are added to a sequence. This leads to a construction in which nonnegativity is much easier to establish.

Let us demonstrate this alternative recursive structure by revisiting Example 3.4. We keep the setting exactly the same as in Chapter 3, but focus only on the conditional masses constructed from B_1 . The focal element B_2 is therefore in the example only for consistency. It contributes to the calculation $Pl(E) = 1$, but it does not enter the individual conditional masses calculated below. To repeat, we have some belief function $Bel(\cdot)$ defined on Ω , we have some event $E \subseteq \Omega$ such that $Bel(E) > 0$ on which we wish to condition, and we are trying to build the conditioned basic belief assignment m' such that

$$Bel(X | E) = \sum_{Y \subseteq X} m'(Y).$$

Example 4.1. In this example, there are two focal sets completely contained in E , namely B_1 and B_2 , and there are two focal sets that overlap both E and $\Omega \setminus E$, namely A_1 and A_2 . Their respective masses are denoted by b_1, b_2, a_1, a_2 . The resulting Venn diagram is shown below.



Since every focal element in this example has a nonempty intersection with E , we have $Pl(E) = b_1 + b_2 + a_1 + a_2 = 1$. Analyzing this example in Example 3.4 led us to find $m'(B_1) = b_1$, $m'(B_1 \cup A'_1) = \frac{b_1}{1-a_1} - b_1$, $m'(B_1 \cup A'_2) = \frac{b_1}{1-a_2} - b_1$, and

$$m'(B_1 \cup A'_1 \cup A'_2) = \frac{b_1}{1-a_1-a_2} - \frac{b_1}{1-a_1} - \frac{b_1}{1-a_2} + b_1.$$

In $m'(B_1 \cup A'_1 \cup A'_2)$, the inclusion-exclusion principle is easily recognizable, and because of it, the nonnegativity of the basic belief assignment is not apparent.

The construction by FH allows us to build up the value for $B_1 \cup A'_1 \cup A'_2$ from the masses assigned to its focal subsets. Jaffray shows us that the math simplifies when we add sets while keeping track of the order in which they came. So recall that FH used an auxiliary function m'' defined on strings of the general form $B_i A'_{j_1} \dots A'_{j_s}$ where $j_1 < \dots < j_s$. This means that they defined $m''(B_1 A'_2 A'_1)$ to be zero. The only strings of nonzero value had to have increasing subscripts in the A' sets. Jaffray now takes away this requirement, and so he defines an auxiliary function $\tilde{m}$ that is defined on ordered sequences of the general form $T = (B, A_1, \dots, A_s)$. The value $m'(X)$ of the conditioned bba is obtained by summing $\tilde{m}$ over all valid ordered sequences whose elements union inside E to form the exact set X . We postpone the exact definition of a valid sequence and of $\mathcal{T}(X)$ to the next section. The goal for now is only to show that retaining both possible orders reproduces the masses derived through FH in Example 3.4. In this simplified setting, Jaffray's construction gives

$$m'(X) = \sum_{T \in \mathcal{T}(X)} \tilde{m}(T),$$

$$\tilde{m}(B) = \frac{b}{Pl(E)},$$

$$\tilde{m}(B, A_1, \dots, A_s) = \tilde{m}(B, A_1, \dots, A_{s-1}) \times \frac{a_s}{Pl(E) - a_1 - \dots - a_s}.$$

Let us look at this from the setting of Example 4.1.

First consider $m'(B)$. Note that $Pl(E) = 1$, and there is only one sequence representing the set B , so

$$m'(B) = \tilde{m}(B) = b.$$

Consider next $m'(B \cup A'_1)$. Again, there is only one sequence of shape $(B, A_1, \dots, A_s)$ representing $B \cup A'_1$, hence we find

$$m'(B \cup A'_1) = \tilde{m}(B, A_1) = b \times \frac{a_1}{1 - a_1} = \frac{b}{1 - a_1} - b.$$

Finally, consider $m'(B \cup A'_1 \cup A'_2)$. Now there are two distinct sequences representing the set $B \cup A'_1 \cup A'_2$, namely the sequence (B, A_1, A_2) and the sequence (B, A_2, A_1) . Their respective weights are

$$\tilde{m}(B, A_1, A_2) = b \times \frac{a_1}{1 - a_1} \times \frac{a_2}{1 - a_1 - a_2}, \text{ and } \tilde{m}(B, A_2, A_1) = b \times \frac{a_2}{1 - a_2} \times \frac{a_1}{1 - a_1 - a_2}.$$

Summing both possible orders gives

$$\begin{aligned} m'(B \cup A'_1 \cup A'_2) &= \tilde{m}(B, A_1, A_2) + \tilde{m}(B, A_2, A_1) \\ &= b \times \frac{a_1}{1 - a_1} \times \frac{a_2}{1 - a_1 - a_2} + b \times \frac{a_2}{1 - a_2} \times \frac{a_1}{1 - a_1 - a_2} \\ &= \frac{ba_1}{1 - a_1} \left(\frac{(1 - a_1) - (1 - a_1 - a_2)}{1 - a_1 - a_2} \right) + \frac{ba_2}{1 - a_2} \left(\frac{(1 - a_2) - (1 - a_1 - a_2)}{1 - a_1 - a_2} \right) \\ &= \frac{ba_1}{1 - a_1 - a_2} - \frac{ba_1}{1 - a_1} + \frac{ba_2}{1 - a_1 - a_2} - \frac{ba_2}{1 - a_2} \\ &= \frac{b(a_1 + a_2)}{1 - a_1 - a_2} - \frac{ba_1}{1 - a_1} - \frac{ba_2}{1 - a_2} \\ &= \frac{b(1 - (1 - a_1 - a_2))}{1 - a_1 - a_2} - \frac{b(1 - (1 - a_1))}{1 - a_1} - \frac{b(1 - (1 - a_2))}{1 - a_2} \\ &= \left(\frac{b}{1 - a_1 - a_2} - b \right) - \left(\frac{b}{1 - a_1} - b \right) - \left(\frac{b}{1 - a_2} - b \right) \\ &= \frac{b}{1 - a_1 - a_2} - \frac{b}{1 - a_1} - \frac{b}{1 - a_2} + b. \end{aligned}$$

So to conclude this example we see that this alternate construction by Jaffray recovers the same values as the construction by FH. One difference is that Jaffray separates the contribution over the two possible orders rather than placing the full inclusion-exclusion expression inside one auxiliary term. Another difference is that nonnegativity of $\tilde{m}$ values is easily verifiable, as it is the product of nonnegative factors.

4.2. Jaffray's auxiliary function

We now introduce Jaffray's auxiliary function and the conditioned bba obtained from it [3]. To maintain consistency with the established theory of evidence, we will present Jaffray's proof using standard Dempster-Shafer notation (Bel, Pl, m) rather than the probabilistic notations he used. Suppose Bel is a belief function defined on 2^Ω , where Ω

is the finite outcome space, and Pl is the associated plausibility function. Take $\mathcal{L}$ to be the set of all probability measures on $(\Omega, 2^\Omega)$. Let m be the Möbius transform of Bel :

$$m(X) = \sum_{Y \subseteq X} (-1)^{|X \setminus Y|} Bel(Y), \quad \forall X \in 2^\Omega.$$

Recall that the Möbius transform uniquely characterizes Bel , and that by definition, m is a basic belief assignment. Let $\mathcal{P}$ be the collection of probability measures consistent with Bel ; $\mathcal{P} = \{P \in \mathcal{L} : P \geq Bel\}$. Let $E \subseteq \Omega$ such that $Bel(E) > 0$ be the event we wish to condition on. As established, we denote the set of focal elements enclosed within E as $\mathcal{B}$, and the set of focal elements that overlap both E and its complement $\Omega \setminus E$ as $\mathcal{A}$.

We once again adopt the notation from FH, so that we have focal sets $B_1, \dots, B_t, A_1, \dots, A_n$, with corresponding masses $b_1, \dots, b_t, a_1, \dots, a_n$. We call

$$T = (B, A_{j_1}, A_{j_2}, \dots, A_{j_k}), \quad 0 \leq k \leq n,$$

a valid ordered sequence when $B \in \mathcal{B}$ and $A_{j_1}, \dots, A_{j_k}$ are distinct elements of $\mathcal{A}$. The case $k = 0$ gives the base sequence $T = (B)$. Writing $b = m(B)$ and $a_{j_x} = m(A_{j_x})$, define the auxiliary function $\tilde{m}$ recursively by

$$\tilde{m}(B) = \frac{b}{Pl(E)},$$

and, for $k \geq 1$,

$$\tilde{m}(B, A_{j_1}, \dots, A_{j_k}) = \tilde{m}(B, A_{j_1}, \dots, A_{j_{k-1}}) \cdot \frac{a_{j_k}}{Pl(E) - \sum_{x=1}^k a_{j_x}}.$$

Every factor in this recursion is nonnegative. In particular, since the mass of B and the masses of all selected A_{j_x} contribute to $Pl(E)$, we have

$$Pl(E) - \sum_{x=1}^k a_{j_x} \geq m(B) = b > 0.$$

Hence every denominator is positive, and consequently $\tilde{m}(T) \geq 0$ for every valid sequence T .

For $X \subseteq E$, let $\mathcal{T}(X)$ denote the collection of all valid sequences that construct the exact set X :

$$\mathcal{T}(X) = \left\{ T = (B, A_{j_1}, \dots, A_{j_s}) \mid T \text{ is valid and } B \cup \left(\bigcup_{\ell=1}^s A'_{j_\ell} \right) = X \right\}.$$

Finally, define

$$\mathcal{P}^E = \{P(\cdot \mid E) : P \in \mathcal{P}\}.$$

Following Jaffray, we view the elements of $\mathcal{P}^E$ as probability measures on the outcome space E , so that FH conditional belief admits the conditioned lower-envelope representation

$$Bel(X \mid E) = \min\{Q(X) : Q \in \mathcal{P}^E\}, \quad X \subseteq E.$$

This restriction to E does not limit the proof. As discussed in Chapter 3, for every $X \subseteq \Omega$ we have $Bel(X \mid E) = Bel(X \cap E \mid E)$, so every conditional belief can be evaluated through a subset of E . It therefore suffices to construct the conditioned bba on 2^E . As shown in Chapter 3, this conditioned lower-envelope representation is equivalent to the FH closed form

$$Bel(X \mid E) = \frac{Bel(X)}{Bel(X) + Pl(E \setminus X)}.$$

Now we have formally defined all concepts necessary to pose Jaffray's theorem.

Theorem 4.1. *Let Bel be a belief function defined on 2^Ω , and let $E \subseteq \Omega$ satisfy $Bel(E) > 0$. Then $Bel(\cdot \mid E) : 2^E \rightarrow [0, 1]$ is a belief function. Its Möbius transform $m' : 2^E \rightarrow [0, 1]$ is the basic belief assignment expressed by*

$$m'(X) = \sum_{T \in \mathcal{T}(X)} \tilde{m}(T), \quad X \subseteq E.$$

4.3. Proof by Jaffray

The proof in this section follows Jaffray [3], with notation adapted for consistency throughout the thesis. As defined, m' is the Möbius transform of $Bel(\cdot \mid E)$ if and only if

$$Bel(X \mid E) = \sum_{Y \subseteq X} m'(Y) = \sum_{Y \subseteq X} \sum_{T \in \mathcal{T}(Y)} \tilde{m}(T), \quad \forall X \subseteq E,$$

which, by the FH closed form, is equivalent to

$$\frac{Bel(X)}{Bel(X) + Pl(E \setminus X)} = \sum_{Y \subseteq X} \sum_{T \in \mathcal{T}(Y)} \tilde{m}(T), \quad \forall X \subseteq E. \quad (4.1)$$

So to prove Theorem 4.1, all we need to show is that Equation 4.1 is upheld for any $X \subseteq E$.

Fix therefore some $X \subseteq E$. Note that if B is a focal element such that $B \subseteq X \subseteq E$, then $B \in \{B_1, \dots, B_t\}$. Next define $\mathcal{B}(X) := \{B \in \mathcal{B} : B \subseteq X\}$; this allows easy notation of all sets B fully contained in set X . Next, define $\mathcal{S}(\emptyset)$ as the set of all finite sequences of distinct events of $\mathcal{A}(X) = \{A \in \mathcal{A} : A \cap E \subseteq X\}$. Then, because $Y = B \cup [(A_1 \cup A_2 \cup \dots \cup A_k) \cap E] \subseteq X$ if and only if $B \subseteq X$ and $A_\ell \cap E \subseteq X$, for $1 \leq \ell \leq k$, we can rewrite the right-hand side of Equation 4.1 to

$$\sum_{B \in \mathcal{B}(X)} \sum_{(A_1, A_2, \dots, A_k) \in \mathcal{S}(\emptyset)} \tilde{m}(B, A_1, A_2, \dots, A_k).$$

If $\mathcal{B}(X) = \emptyset$, then $Bel(X) = 0$ and the displayed sum is empty, so Equation 4.1 holds immediately. We may therefore assume from now on that $\mathcal{B}(X) \neq \emptyset$.

Write $L := |\mathcal{A}(X)|$. We will decompose this sum further by creating a notation for subsets of $\mathcal{S}(\emptyset)$ with a common initial sequence of length $1 \leq \ell \leq L$:

$$\mathcal{S}(A_{j_1}, A_{j_2}, \dots, A_{j_\ell}) = \{(\overline{A_{j_1}}, \overline{A_{j_2}}, \dots, \overline{A_{j_k}}) \in \mathcal{S}(\emptyset) : k \geq \ell \text{ and } \overline{A_{j_m}} = A_{j_m}, 1 \leq m \leq \ell\}.$$

We then denote the corresponding partial summation of weights for a given $B_i \in \mathcal{B}(X)$:

$$S(B_i, A_{j_1}, A_{j_2}, \dots, A_{j_\ell}) = \sum_{(\overline{A_{j_1}}, \overline{A_{j_2}}, \dots, \overline{A_{j_k}}) \in \mathcal{S}(A_{j_1}, A_{j_2}, \dots, A_{j_\ell})} \tilde{m}(B_i, \overline{A_{j_1}}, \overline{A_{j_2}}, \dots, \overline{A_{j_k}}).$$

For the empty initial sequence, define

$$S(B_i) = \sum_{(\overline{A_{j_1}}, \dots, \overline{A_{j_k}}) \in \mathcal{S}(\emptyset)} \tilde{m}(B_i, \overline{A_{j_1}}, \dots, \overline{A_{j_k}}),$$

where the empty sequence $k = 0$ contributes the term $\tilde{m}(B_i)$. Thus, Equation 4.1 becomes

$$\frac{Bel(X)}{Bel(X) + Pl(E \setminus X)} = \sum_{B_i \in \mathcal{B}(X)} S(B_i).$$

We shall now find an explicit expression for $S(B_i)$ by induction.

Let $(A_1, A_2, \dots, A_L)$ be any given ordering of $\mathcal{A}(X)$, and let B be a given event of $\mathcal{B}(X)$. Although we write an initial sequence as $(A_1, \dots, A_\ell)$, this notation represents an arbitrary ordered prefix of length ℓ . After fixing such a prefix, the remaining elements of $\mathcal{A}(X)$ are relabelled $A_{\ell+1}, \dots, A_L$. Thus, a statement at order ℓ is understood to hold for every valid ordered prefix of that length. Every sequence of events in $\mathcal{S}(A_1, A_2, \dots, A_\ell)$, $0 \leq \ell \leq L - 1$, except $(A_1, A_2, \dots, A_\ell)$, has at least an $(\ell + 1)$ -th term, A_j , where $\ell + 1 \leq j \leq L$; therefore

$$S(B, A_1, A_2, \dots, A_\ell) = \tilde{m}(B, A_1, A_2, \dots, A_\ell) + \sum_{j=\ell+1}^L S(B, A_1, A_2, \dots, A_\ell, A_j), \quad \text{for } 0 \leq \ell \leq L-1.$$

We now prove by backward induction that for every valid ordered prefix of length $0 \leq \ell \leq L$,

$$S(B, A_1, \dots, A_\ell) = \tilde{m}(B, A_1, \dots, A_\ell) \times \frac{Pl(E) - \sum_{x=1}^{\ell} a_x}{Pl(E) - \sum_{x=1}^L a_x}.$$

For $\ell = L$, the set $\mathcal{S}(A_1, \dots, A_L)$ contains only the sequence $(A_1, \dots, A_L)$. Hence

$$S(B, A_1, \dots, A_L) = \tilde{m}(B, A_1, \dots, A_L),$$

so the base case holds.

Suppose now that the formula holds for every valid ordered prefix of length $\ell + 1$, where $0 \leq \ell \leq L - 1$. Recall the recursive definition

$$\tilde{m}(B, A_1, \dots, A_k) = \tilde{m}(B, A_1, \dots, A_{k-1}) \cdot \frac{a_k}{Pl(E) - \sum_{x=1}^k a_x}.$$

We can roll out the recursion so that for general sequence $T = (B, A_1, \dots, A_k)$, we have

$$\tilde{m}(T) = \frac{b \times a_1 \times a_2 \times \dots \times a_k}{Pl(E) \times [Pl(E) - a_1] \times [Pl(E) - a_1 - a_2] \times \dots \times [Pl(E) - \sum_{x=1}^k a_x]}.$$

Let $T_\ell = (B, A_1, \dots, A_\ell)$ and $T_m = (B, A_1, \dots, A_\ell, A_m)$. We then have

$$S(B, A_1, \dots, A_\ell) = \tilde{m}(B, A_1, \dots, A_\ell) + \sum_{m=\ell+1}^L S(B, A_1, \dots, A_\ell, A_m)$$

$$= \tilde{m}(T_\ell) + \sum_{m=\ell+1}^L S(T_m)$$

which by inductive hypothesis gives

$$= \tilde{m}(T_\ell) + \sum_{m=\ell+1}^L \tilde{m}(T_m) \times \frac{Pl(E) - \sum_{x=1}^\ell a_x - a_m}{Pl(E) - \sum_{x=1}^L a_x}$$

here we can substitute the recursive formula for $\tilde{m}(T_m)$ to get

$$\begin{aligned} &= \tilde{m}(T_\ell) + \sum_{m=\ell+1}^L \tilde{m}(T_\ell) \times \frac{a_m}{Pl(E) - \sum_{x=1}^\ell a_x - a_m} \times \frac{Pl(E) - \sum_{x=1}^\ell a_x - a_m}{Pl(E) - \sum_{x=1}^L a_x} \\ &= \tilde{m}(T_\ell) + \sum_{m=\ell+1}^L \tilde{m}(T_\ell) \times \frac{a_m}{Pl(E) - \sum_{x=1}^L a_x} \\ &= \tilde{m}(B, A_1, \dots, A_\ell) \left[1 + \sum_{m=\ell+1}^L \frac{a_m}{Pl(E) - \sum_{x=1}^L a_x} \right]. \end{aligned}$$

Finally, rewriting

$$1 = \frac{Pl(E) - \sum_{x=1}^L a_x}{Pl(E) - \sum_{x=1}^L a_x},$$

and noting that

$$-\sum_{x=1}^L a_x + \sum_{m=\ell+1}^L a_m = -\sum_{x=1}^\ell a_x,$$

then gives us

$$S(B, A_1, \dots, A_\ell) = \tilde{m}(B, A_1, \dots, A_\ell) \times \frac{Pl(E) - \sum_{x=1}^\ell a_x}{Pl(E) - \sum_{x=1}^L a_x}.$$

This shows that if the formula holds for every valid ordered prefix of length $\ell + 1$, then it also holds for every valid ordered prefix of length ℓ . By backward induction, the explicit formulation of $S(B, A_1, \dots, A_\ell)$ therefore holds for every $0 \leq \ell \leq L$.

In particular, for $\ell = 0$, we then have

$$S(B) = \tilde{m}(B) \times \frac{Pl(E)}{Pl(E) - \sum_{j=1}^L a_j} = \frac{b}{Pl(E) - \sum_{j=1}^L a_j}.$$

To complete the proof, it remains only to compare our explicit formulation,

$$\sum_{B \in \mathcal{B}(X)} S(B) = \frac{\sum_{B \in \mathcal{B}(X)} m(B)}{Pl(E) - \sum_{j=1}^L a_j},$$

with the FH closed form,

$$\frac{Bel(X)}{Bel(X) + Pl(E \setminus X)}.$$

It is straightforward that the numerator is equal to $Bel(X)$, since by definition $\mathcal{B}(X)$ contains all focal elements completely contained in X :

$$\sum_{B \in \mathcal{B}(X)} m(B) = Bel(X).$$

For the denominator, recall that $\mathcal{A}(X) = \{A \in \mathcal{A} : A \cap E \subseteq X\}$. We can express the belief of $X \cup E^c$ by separating the focal elements contained in $X \cup E^c$:

$$\begin{aligned} Bel(X \cup E^c) &= \sum_{Y \subseteq X \cup E^c} m(Y) \\ &= \sum_{Y \subseteq X} m(Y) + \sum_{Y \subseteq E^c} m(Y) + \sum_{j=1}^L m(A_j) \\ &= Bel(X) + Bel(E^c) + \sum_{j=1}^L m(A_j). \end{aligned}$$

Thus, substituting this rearranged sum into the denominator yields:

$$\begin{aligned} Pl(E) - \sum_{j=1}^L a_j &= Pl(E) - [Bel(X \cup E^c) - Bel(E^c) - Bel(X)] \\ &= (1 - Bel(E^c)) - Bel(X \cup E^c) + Bel(E^c) + Bel(X) \\ &= 1 - Bel(X \cup E^c) + Bel(X) \\ &= Pl(E \setminus X) + Bel(X). \end{aligned}$$

So we have shown that m' is the Möbius transform of $Bel(\cdot \mid E)$. Taking $X = \emptyset$ in the established identity gives $m'(\emptyset) = 0$, while taking $X = E$ gives

$$\sum_{Y \subseteq E} m'(Y) = Bel(E \mid E) = 1.$$

As established, $\tilde{m}(T) \geq 0$ for every valid ordered sequence T . Since $m'(X)$ is a sum of such sequence weights, it follows that $m'(X) \geq 0$ for every $X \subseteq E$. Therefore, m' is a valid basic belief assignment, and $Bel(\cdot | E)$ is a valid belief function.

4.4. Worked example

Some of the inner mechanisms of Jaffray's proof are hard to grasp in all generality. We therefore give a worked example constructed for this thesis in which every stage of Jaffray's proof can be calculated explicitly. The lowest level of construction is Jaffray's auxiliary function $\tilde{m}$, which assigns positive weights to ordered sequences and is defined recursively. Jaffray constructs the bba $m'(X)$ by summing the value of $\tilde{m}$ over all ordered sequences whose union inside E equals X . To evaluate the sum $Bel(X | E) = \sum_{Y \subseteq X} m'(Y)$, Jaffray reorganizes the sequence weights into a partial sum S , for which both a recursive and an explicit definition are presented. This avoids the need to calculate every individual value of m' . We now review all these steps in the following example. Let

$$\Omega = \{x_1, x_2, x_3, x_4, x_5, x_6, x_7\} \quad E = \{x_1, x_2, x_3, x_4, x_7\},$$

and let the original belief function be defined by the bba

$$\begin{array}{llllll} B_1 = \{x_1\} & B_2 = \{x_7\} & A_1 = \{x_2, x_5\} & A_2 = \{x_2, x_3, x_6\} & A_3 = \{x_4, x_5, x_6\} & C = \{x_5, x_6\} \\ m(B_1) = \frac{1}{16} & m(B_2) = \frac{1}{8} & m(A_1) = \frac{1}{4} & m(A_2) = \frac{3}{16} & m(A_3) = \frac{1}{8} & m(C) = \frac{1}{4} \end{array}$$

Note the labeling of sets such that the sets $\mathcal{B} = \{B_1, B_2\}$ are enclosed in E , the sets $\mathcal{A} = \{A_1, A_2, A_3\}$ partially overlap E , and the set C falls completely outside of E . Therefore C is the only focal element that does not contribute to $Pl(E)$, thus we note $Pl(E) = 1 - m(C) = \frac{3}{4}$.

We look first at Jaffray's auxiliary function $\tilde{m}$. For the base ordered sequence $T = (B_1)$ the auxiliary function just normalizes the bba value. So

$$\tilde{m}(B_1) = \frac{m(B_1)}{Pl(E)} = \frac{1/16}{3/4} = \frac{1}{12}.$$

Next, we can extend this sequence by the partially overlapping focal element A_1 . Here we use Jaffray's recursive definition of the auxiliary function. We see

$$\tilde{m}(B_1, A_1) = \tilde{m}(B_1) \frac{m(A_1)}{Pl(E) - m(A_1)} = \frac{1}{12} \frac{1/4}{3/4 - 1/4} = \frac{1}{24}.$$

Recall that this specific sequence refers to the set $B_1 \cup (A_1 \cap E) = \{x_1, x_2\}$. To calculate $m'(\{x_1, x_2\})$, we see that no other sequence constructs the same set, hence $m'(\{x_1, x_2\}) = 1/24$.

The reason why $\tilde{m}$ is defined on ordered sequences becomes clear when a second partially overlapping focal element A_2 is added. Namely, processing A_1 before A_2 gives

$$\tilde{m}(B_1, A_1, A_2) = \tilde{m}(B_1, A_1) \frac{m(A_2)}{Pl(E) - m(A_1) - m(A_2)} = \frac{1}{24} \frac{3/16}{3/4 - 1/4 - 3/16} = \frac{1}{40},$$

while reversing this order gives

$$\tilde{m}(B_1, A_2, A_1) = \tilde{m}(B_1, A_2) \frac{m(A_1)}{Pl(E) - m(A_2) - m(A_1)} = \frac{1}{36} \frac{1/4}{3/4 - 3/16 - 1/4} = \frac{1}{45},$$

showing that $\tilde{m}(B_1, A_1, A_2) \neq \tilde{m}(B_1, A_2, A_1)$.

Note here what this sensitivity to order achieves. Namely, $\tilde{m}$ is a recursive product of nonnegative factors, thus enforcing nonnegativity of m' by construction. Contrast this with the auxiliary function used by FH, which caused an inclusion-exclusion pattern that made nonnegativity hard to prove.

The different ordered sequences necessarily construct the same set, namely $B_1 \cup ((A_1 \cup A_2) \cap E) = \{x_1, x_2, x_3\}$. Note then that

$$A_1 \cap E = \{x_2\} \subseteq A_2 \cap E = \{x_2, x_3\},$$

implying that the sequence $T = (B_1, A_2)$ also constructs this set. No other sequences construct the same set, hence we can calculate the resulting bba value

$$\begin{aligned} m'(\{x_1, x_2, x_3\}) &= \tilde{m}(B_1, A_2) + \tilde{m}(B_1, A_1, A_2) + \tilde{m}(B_1, A_2, A_1) \\ &= \frac{1}{36} + \frac{1}{40} + \frac{1}{45} = \frac{3}{40}. \end{aligned}$$

This calculation illustrates the relation between $\tilde{m}$ and m' , and so we can now continue our example by discussing Jaffray's S function. For this we need the remaining values of m' which follow from the same recursion. Since $m(B_2) = 2m(B_1)$, every sequence beginning with B_2 has twice the weight of the corresponding sequence beginning with B_1 . The resulting values are

Y	$m'(Y)$	Y	$m'(Y)$
$\{x_1\}$	$\frac{1}{12}$	$\{x_7\}$	$\frac{1}{6}$
$\{x_1, x_2\}$	$\frac{1}{24}$	$\{x_2, x_7\}$	$\frac{1}{12}$
$\{x_1, x_4\}$	$\frac{1}{60}$	$\{x_4, x_7\}$	$\frac{1}{30}$
$\{x_1, x_2, x_3\}$	$\frac{3}{40}$	$\{x_2, x_3, x_7\}$	$\frac{3}{20}$
$\{x_1, x_2, x_4\}$	$\frac{1}{40}$	$\{x_2, x_4, x_7\}$	$\frac{1}{20}$
$\{x_1, x_2, x_3, x_4\}$	$\frac{11}{120}$	$\{x_2, x_3, x_4, x_7\}$	$\frac{11}{60}$

All other subsets have zero mass, and the values in the table sum to one. This then completely maps out the resulting conditional belief function, but it remains to verify that the masses indeed generate FH conditional belief.

So say we wish to know the conditional belief in the set $X = \{x_1, x_2, x_3, x_7\}$ given E . Both our interior focal elements lie in X . However, of the partially overlapping focal elements, only A_1 and A_2 have their intersections with E contained in X . Hence, using Jaffray's notations we have $\mathcal{B}(X) = \{B_1, B_2\}$, and $\mathcal{A}(X) = \{A_1, A_2\}$. Recall then that for the general ordered sequence $T = (B, A_{j_1}, A_{j_2}, \dots, A_{j_k})$, the partial sum $S(T)$ is the sum over the $\tilde{m}$ values of all possible sequences that have T as starting sequence. From this point onward, S is evaluated relative to the fixed target set X , so only continuations by elements of $\mathcal{A}(X) = \{A_1, A_2\}$ are included. So, we see

$$S(B_1, A_1) = \tilde{m}(B_1, A_1) + S(B_1, A_1, A_2).$$

The sequence $T = (B_1, A_1, A_2)$ then has no further continuation, so that $S(B_1, A_1, A_2) = \tilde{m}(B_1, A_1, A_2)$.

Consequently,

$$S(B_1, A_1) = \frac{1}{24} + \frac{1}{40} = \frac{1}{15},$$

and we similarly see

$$S(B_1, A_2) = \tilde{m}(B_1, A_2) + \tilde{m}(B_1, A_2, A_1) = \frac{1}{36} + \frac{1}{45} = \frac{1}{20}.$$

So we can then calculate the base level partial sum

$$S(B_1) = \tilde{m}(B_1) + S(B_1, A_1) + S(B_1, A_2) = \frac{1}{12} + \frac{1}{15} + \frac{1}{20} = \frac{1}{5}.$$

This showcases the role that S plays. It is a simple recursion that enables us to easily sum the values of $\tilde{m}$ over all possible ordered sequences. Now Jaffray's induction shows that this S has the alternative explicit formulation

$$S(B) = \tilde{m}(B) \times \frac{Pl(E)}{Pl(E) - \sum_{A \in \mathcal{A}(X)} m(A)} = \frac{m(B)}{Pl(E) - \sum_{A \in \mathcal{A}(X)} m(A)}.$$

For B_1 , this gives

$$S(B_1) = \frac{1/16}{3/4 - 1/4 - 3/16} = \frac{1}{5},$$

which indeed corresponds with our full calculation. So now we have exemplified how S is constructed recursively, and we have confirmed that its explicit formulation reaches the same result. Once those results were generally established, the remainder of Jaffray's proof was spent on showing two equivalences, namely

$$\sum_{Y \subseteq X} m'(Y) = \sum_{B \in \mathcal{B}(X)} S(B) = \frac{Bel(X)}{Bel(X) + Pl(E \setminus X)}.$$

Let us confirm these results in our example by calculating the three values. First, using the calculated m' values, we confirm that

$$\begin{aligned}
\sum_{Y \subseteq X} m'(Y) &= m'(\{x_1\}) + m'(\{x_7\}) + m'(\{x_1, x_2\}) + m'(\{x_2, x_7\}) \\
&\quad + m'(\{x_1, x_2, x_3\}) + m'(\{x_2, x_3, x_7\}) \\
&= \frac{1}{12} + \frac{1}{6} + \frac{1}{24} + \frac{1}{12} + \frac{3}{40} + \frac{3}{20} = \frac{3}{5}.
\end{aligned}$$

Next, we consider $\sum_{B \in \mathcal{B}(X)} S(B)$. Since $m(B_2) = 2m(B_1)$, it turns out that $S(B_2) = 2S(B_1) = 2/5$. Hence it follows that

$$\sum_{B \in \mathcal{B}(X)} S(B) = S(B_1) + S(B_2) = \frac{3}{5}.$$

Finally we calculate FH conditional belief using the FH closed form. Note that $Pl(E \setminus X) = m(A_3) = 1/8$, and note furthermore that $Bel(X) = m(B_1) + m(B_2) = 1/16 + 1/8 = 3/16$. This then gives us

$$\frac{Bel(X)}{Bel(X) + Pl(E \setminus X)} = \frac{3/16}{3/16 + 1/8} = \frac{3}{5}.$$

So indeed this confirms the final two equivalences in the proof in our example. This example makes the structure in Jaffray's proof explicit. The auxiliary function $\tilde{m}$ assigns nonnegative weights to individual ordered sequences, while the conditioned bba m' sums the $\tilde{m}$ values over all sequences producing the same set. The partial sum S is then a tool to calculate the final belief. So evaluating S adds values of $\tilde{m}$, allowing the corresponding conditional belief to be calculated without first determining every value of m' . In the present example, all three calculations produce the same result $3/5$, confirming concretely that Jaffray's ordered construction generates FH conditional belief. The ordered construction established here forms the starting point for Chapter 5, where we investigate where order enters the construction of conditional mass.

5. The sequential structure of conditional mass

The overarching progression we have seen going from Chapter 3 to Chapter 4 is that moving from unordered strings to ordered sequences makes proving the central theorem simpler. In this chapter, we investigate where order enters the construction of conditional mass. We begin by looking at the operation of successive FH conditioning, noting that sequential conditioning is noncommutative. We then compare this dependence on order to the ordered sequences on which Jaffray's auxiliary function $\tilde{m}$ is defined. These are two distinct operations, as the first operation concerns successive conditioning on events, while the second concerns the ordering of focal elements within the construction of conditional mass. We nevertheless find a similarity between the operations. Namely, under the assumption that partially overlapping focal sets are mutually exclusive, both operations can be interpreted as a reduction in plausibility. This insight leads to an adapted version of Jaffray's proof under the assumption of mutual exclusivity. Next we introduce a mathematical construction developed in this thesis to remove the assumption of mutual exclusivity, which then leads to a general adaptation of Jaffray's proof. Both adaptations are intended to provide a more intuitive perspective on Jaffray's construction. Finally, we return to the worked example from Section 4.4 to show how the general proof produces the same results as Jaffray's proof.

The noncommutativity of successive FH conditioning and Jaffray's construction are results from the literature. The comparison between these forms of order, and their interpretation through changes in plausibility are developed in this thesis. These constructions are used to reformulate Jaffray's proof, so they are fundamentally not different proofs.

5.1. Noncommutativity and ordered mass

For classical probabilities, conditioning on an event E_1 and then E_2 yields exactly the same probability measure as conditioning on E_2 then E_1 . This means that $P((X | E_1) | E_2) = P((X | E_2) | E_1)$, which as a property is called commutativity. Through an example we can illustrate that this property does not hold for successive FH conditioning. Namely, consider a belief function on $\Omega = \{1, 2, 3, 4\}$ with the basic belief assignments

$$m(\{2\}) = 0.2, \quad m(\{1, 2\}) = 0.2, \quad m(\{3, 4\}) = 0.4, \quad m(\{1, 4\}) = 0.2.$$

Suppose then that we have two sets of evidence, $E_1 = \{1, 2, 3\}$ and $E_2 = \{2, 3, 4\}$, and we wish to evaluate our final conditional belief in the set $\{2\}$. If we condition on E_1

first, we calculate

$$Bel(\{2\} \mid E_1) = \frac{Bel(\{2\} \cap E_1)}{Bel(\{2\} \cap E_1) + Pl(E_1 \setminus \{2\})} = \frac{Bel(\{2\})}{Bel(\{2\}) + Pl(\{1, 3\})} = \frac{0.2}{0.2 + 0.8} = 0.2.$$

To condition on E_2 afterwards, we need to first calculate $Pl(\{3, 4\} \mid E_1)$

$$= 1 - Bel(\{1, 2\} \mid E_1) = 1 - \frac{Bel(\{1, 2\} \cap E_1)}{Bel(\{1, 2\} \cap E_1) + Pl(E_1 \setminus \{1, 2\})} = 1 - \frac{0.4}{0.4 + 0.4} = 0.5,$$

this then gives us

$$Bel((\{2\} \mid E_1) \mid E_2) = \frac{Bel(\{2\} \mid E_1)}{Bel(\{2\} \mid E_1) + Pl(\{3, 4\} \mid E_1)} = \frac{0.2}{0.2 + 0.5} = \frac{2}{7}.$$

Performing the same calculation in the opposite order of conditioning, one finds $Bel((\{2\} \mid E_2) \mid E_1) = \frac{3}{11}$. Since $\frac{2}{7} > \frac{3}{11}$, this confirms that successive FH conditioning is noncommutative. This example thereby demonstrates the dependence on order for successive FH conditioning.

Recall that FH defined their auxiliary function m'' on unordered strings representing unions of focal elements. Such a union of sets is agnostic to the order in which the sets are grouped together. Consequently, contributions from different orderings are grouped together, and this specifically led to the inclusion-exclusion structure in their recursive definition of m'' . Jaffray instead defines his auxiliary function $\tilde{m}$ on ordered sequences. The ordering of the focal elements is therefore retained explicitly, and generally influences the value of $\tilde{m}$. The recursive definition Jaffray found for $\tilde{m}$ is simpler than its counterpart in FH, and more importantly, it retains nonnegativity by construction. This all suggests that order plays an important structural role in the creation of masses underlying $Bel(\cdot \mid E)$. There is then a conceptual parallel between successive FH conditioning, and the construction of Jaffray's auxiliary function $\tilde{m}$, as both operations are sensitive to order.

The order dependence of $\tilde{m}$ should nevertheless be distinguished from the noncommutativity of successive conditioning. Recall namely that the value of $Bel(X \mid E_1)$ is obtained by summing the values of $\tilde{m}$ over the ordered sequences whose constructed sets are contained in X . Thus, a single evaluation of conditional belief may require the aggregation of many different ordered sequences. If the resulting belief function is subsequently conditioned on E_2 , then an entirely different auxiliary function must be constructed from the conditioned bba corresponding to $Bel(\cdot \mid E_1)$. In short, both sequential conditioning of belief and the evaluation of $\tilde{m}$ are dependent on some sort of sequencing of events, but the two orders concern different mathematical objects. Keeping this distinction in mind, the next section investigates whether their recursive structures can nevertheless be related.

5.2. Comparing operations

Suppose for a moment that we are conditioning on an event E where $Bel(E) > 0$, and let m be the bba corresponding to Bel . However, suppose we do not know all of the

focal elements of Bel . We are currently only aware of the focal elements B_1 and A_1 with $B_1 \subset E$, $A_1 \cap E \neq \emptyset$, and $A_1 \cap (\Omega \setminus E) \neq \emptyset$, with corresponding masses $m(B_1)$ and $m(A_1)$. Let $\tilde{m}_E$ denote Jaffray's auxiliary function used to construct the conditioned bba. Although we do not know the remaining focal elements individually, we already have enough information to evaluate $\tilde{m}_E(B_1, A_1)$. Namely, Jaffray's recurrence gives

$$\tilde{m}_E(B_1, A_1) = \frac{m(B_1)}{Pl(E)} \times \frac{m(A_1)}{Pl(E) - m(A_1)}.$$

The important feature of this expression is its locality. Once $Pl(E)$ is known, the value of $\tilde{m}_E(B_1, A_1)$ depends only on the masses assigned to B_1 and A_1 . The remaining focal elements influence this value only through $Pl(E)$, so once $Pl(E)$ is known, their individual sets and masses are irrelevant. This locality is what allows Jaffray to construct conditional mass recursively. Given that $m(B_1)$, $m(A_1)$, and $Pl(E)$ are already known, a new focal element like A_2 can be added to the sequence as soon as we know $m(A_2)$. Thus, each new focal element contributes one additional factor, determined by its own mass. The value of $\tilde{m}$ can therefore be built one step at a time, without first reconstructing the complete conditioned bba.

This locality also means that the conditioning event itself matters only through the quantities appearing in the recurrence. Namely, suppose that for the same belief function, E' is another event that encloses B_1 , partially overlaps A_1 and preserves the value $Pl(E) = Pl(E')$. Then the corresponding $\tilde{m}$ from conditioning on that set results in the same value; $\tilde{m}_E(B_1, A_1) = \tilde{m}_{E'}(B_1, A_1)$. We will use this specific observation in Section 5.3.

Let us now extend this example by introducing another focal element A_2 that partially overlaps E and has mass $m(A_2)$. Assume that $m(B_1) + m(A_1) + m(A_2) < 1$, so that further focal elements remain unspecified. Jaffray's recurrence then gives

$$\tilde{m}_E(B_1, A_1, A_2) = \frac{m(B_1)}{Pl(E)} \times \frac{m(A_1)}{Pl(E) - m(A_1)} \times \frac{m(A_2)}{Pl(E) - m(A_1) - m(A_2)}.$$

For the remainder of this comparison, assume that all focal elements intersecting E are mutually exclusive. Our aim now is to compare the update from $\tilde{m}_E(B_1, A_1)$ to $\tilde{m}_E(B_1, A_1, A_2)$ with the changes in conditional belief when the set E changes. Towards this aim, define $E_0 = E$, $E_1 = E_0 \setminus A_1$, and $E_2 = E_1 \setminus A_2$. Let $\tilde{m}_{E_1}$ denote Jaffray's auxiliary function for constructing the bba corresponding to $Bel(\cdot \mid E_1)$, and define $\tilde{m}_{E_2}$ analogously. The sequence E_0, E_1, E_2 provides an auxiliary way to remove the contributions of A_1 and A_2 from the plausibility of the conditioning event one at a time. In particular, passing from E_0 to $E_1 = E_0 \setminus A_1$ removes the contribution of A_1 while leaving the contributions of the other focal elements unchanged. So by updating our perspective from E_0 to E_1 we arrive at a state where the ambiguity related to A_1 has been resolved. Jaffray's recurrence then gives the following value:

$$\tilde{m}_{E_1}(B_1, A_2) = \frac{m(B_1)}{Pl(E_1)} \times \frac{m(A_2)}{Pl(E_1) - m(A_2)}.$$

By mutual exclusivity, removing A_1 from E_0 reduces the plausibility by exactly $m(A_1)$, so that $Pl(E_1) = Pl(E_0) - m(A_1)$. Hence, we can rewrite to

$$\tilde{m}_{E_1}(B_1, A_2) = \frac{m(B_1)}{Pl(E_1)} \times \frac{m(A_2)}{Pl(E) - m(A_1) - m(A_2)}.$$

Comparing $\tilde{m}_E(B_1, A_1, A_2)$ with $\tilde{m}_{E_1}(B_1, A_2)$, we see that their final recursive factors are identical. These identical factors allow us to relate the extension of the ordered sequence to the change in plausibility from E_0 to E_1 .

Note then that when A_1 is excluded from E_0 , the plausibility drops by $m(A_1)$, so we see $Pl(E_1) = Pl(E_0) - m(A_1)$. Hence

$$\tilde{m}_{E_0}(B_1, A_1) = \tilde{m}_{E_0}(B_1) \times \frac{m(A_1)}{Pl(E_0) - m(A_1)} = \tilde{m}_{E_0}(B_1) \times \frac{Pl(E_0) - Pl(E_1)}{Pl(E_1)}.$$

To understand what this fraction actually represents, let us consider the conditional belief in B_1 under the conditioning events E_0 and E_1 . The FH closed form then gives $Bel(B_1 | E_k) = \frac{m(B_1)}{Pl(E_k)}$. Hence,

$$\begin{aligned} \frac{Pl(E_0) - Pl(E_1)}{Pl(E_1)} &= \frac{\frac{1}{Pl(E_1)} - \frac{1}{Pl(E_0)}}{\frac{1}{Pl(E_0)}} = \frac{\frac{m(B_1)}{Pl(E_1)} - \frac{m(B_1)}{Pl(E_0)}}{\frac{m(B_1)}{Pl(E_0)}} \\ &= \frac{Bel(B_1 | E_1) - Bel(B_1 | E_0)}{Bel(B_1 | E_0)}. \end{aligned}$$

Therefore, we can express the update in $\tilde{m}$ in terms of the corresponding change in conditional belief:

$$\tilde{m}_{E_0}(B_1, A_1) = \tilde{m}_{E_0}(B_1) \times \frac{Bel(B_1 | E_1) - Bel(B_1 | E_0)}{Bel(B_1 | E_0)}.$$

This fraction has a direct interpretation in terms of the change in conditional belief. The numerator is the increase in conditional belief in B_1 produced by removing A_1 from the conditioning event. Dividing this increase by $Bel(B_1 | E_0)$ gives the relative increase in conditional belief in B_1 . The value $\tilde{m}_{E_0}(B_1, A_1)$ is therefore obtained by multiplying $\tilde{m}_{E_0}(B_1)$ by this relative increase. So Jaffray's ordered sequence of sets is directly tracking how much our belief increases as we sequentially prune away ignorance.

Applying the same logic to $E_2 = E_1 \setminus A_2$, we find

$$\tilde{m}_{E_0}(B_1, A_1, A_2) = \tilde{m}_{E_0}(B_1, A_1) \times \frac{Bel(B_1 | E_2) - Bel(B_1 | E_1)}{Bel(B_1 | E_1)}.$$

So, if we extend the example to many A_i sets, where $E_0 = E, E_1 = E_0 \setminus A_1, E_2 = E_1 \setminus A_2, \dots$, we obtain the recurrence

$$\tilde{m}_{E_0}(B_1, A_1, \dots, A_k) = \tilde{m}_{E_0}(B_1, A_1, \dots, A_{k-1}) \times \frac{Bel(B_1 | E_k) - Bel(B_1 | E_{k-1})}{Bel(B_1 | E_{k-1})},$$

or equivalently

$$\tilde{m}_E(B_1, A_1, \dots, A_k) = \tilde{m}_E(B_1, A_1, \dots, A_{k-1}) \times \frac{Pl(E_{k-1}) - Pl(E_k)}{Pl(E_k)}.$$

What this shows is that the parallel identified in Section 5.1 goes deeper than being an analogy. We can iteratively construct different sets of evidence to condition on in such a way that the successive changes in conditional belief and the recurrence for $\tilde{m}$ are governed by the same ordered reductions in plausibility. More precisely, the relative change in conditional belief we see by moving from E_{k-1} to E_k equals the next multiplicative factor in Jaffray's recurrence. A differently ordered sequence T therefore results in a different value for $\tilde{m}$ as this changes the intermediate plausibility values.

Unfortunately, the recurrences shown above rely on the mutual exclusivity assumption, since otherwise the statement $Pl(E_1) = Pl(E_0) - m(A_1)$ does not hold. Fortunately, we can remove this restriction by changing the perspective to updating the belief functions instead of updating the evidence on which to condition. How exactly this works will be presented in Section 5.4.

5.3. Mutually exclusive adaptation

In this section we present a restricted version of Jaffray's theorem. We will operate under the assumption that all focal elements intersecting the conditioning set are pairwise mutually exclusive. Under this assumption we can utilize the intuitive recursion found in Section 5.2, thereby creating a more intuitive version of Jaffray's proof.

Notation and setup

Let Ω be a finite outcome space, let Bel be some belief function on 2^Ω , and let m be the corresponding basic belief assignment. Let then $E \subseteq \Omega$ be the evidence set we wish to condition on, where we assume $Bel(E) > 0$. Let $\mathcal{B}$ denote the set of focal elements of Bel that are enclosed in E , and let $\mathcal{A}$ denote the set of focal elements of Bel that intersect both E and $\Omega \setminus E$. In this section, we assume that Bel and E are such that all focal elements in $\mathcal{A} \cup \mathcal{B}$ are pairwise mutually exclusive. That is, for any distinct $Y_1, Y_2 \in \mathcal{A} \cup \mathcal{B}$, we assume $Y_1 \cap Y_2 = \emptyset$.

Take $\mathcal{L}$ to be the set of all probability measures on $(\Omega, 2^\Omega)$, and let $\mathcal{P}$ be the collection of probability measures consistent with Bel ; $\mathcal{P} = \{P \in \mathcal{L} : P \geq Bel\}$. For a fixed target set $X \subseteq E$, define $\mathcal{A}(X) = \{A \in \mathcal{A} : A \cap E \subseteq X\}$. Throughout the remainder of this chapter, resolving the ambiguity related to X means removing from $Pl(E)$ the contributions of exactly the focal elements in $\mathcal{A}(X)$. Focal elements in $\mathcal{A} \setminus \mathcal{A}(X)$ may still partially overlap E , but they intersect $E \setminus X$ and therefore remain part of the competing plausibility $Pl(E \setminus X)$. Hence, after removing all ambiguity related to X , there can still be other partially overlapping focal elements in the original belief function. For any ordered sequence $(A_1, \dots, A_k)$ of distinct elements of $\mathcal{A}$, define a sequence of sets $E_0 = E, E_1 = E_0 \setminus A_1, E_2 = E_1 \setminus A_2, \dots$. Conceptually, applying this construction to

the elements of $\mathcal{A}(X)$ removes the ambiguity related to X one focal element at a time, producing a predictable reduction in plausibility at every step. We can then add these focal elements back one by one, where each addition restores a predictable amount of plausibility.

For an ordered sequence of focal elements $T = (B, A_1, A_2, \dots, A_k)$ where $B \in \mathcal{B}$, and the $A_i \in \mathcal{A}$ are all distinct, we define the recursive auxiliary function $\tilde{m}(T)$ by $\tilde{m}(B) = \frac{m(B)}{Pl(E)}$, and

$$\tilde{m}(B, A_1, A_2, \dots, A_k) = \tilde{m}(B, A_1, A_2, \dots, A_{k-1}) \times \frac{Pl(E_{k-1}) - Pl(E_k)}{Pl(E_k)}.$$

Let $\mathcal{T}(X)$ for $X \subseteq \Omega$ denote the set of all ordered sequences $T = (B, A_1, A_2, \dots, A_k)$ that construct the set X :

$$\mathcal{T}(X) = \left\{ T = (B, A_1, \dots, A_k) \mid B \cup \left(\bigcup_{i=1}^k A_i \cap E \right) = X \right\}.$$

Finally, define $\mathcal{P}^E = \{P(\cdot \mid E) \text{ for } P \in \mathcal{P}\}$, so that FH conditional belief admits the conditioned lower-envelope representation

$$Bel(X \mid E) = \min\{Q(X) : Q \in \mathcal{P}^E\}.$$

As established before, for $X \subseteq E$ this is equivalent to the FH closed form

$$Bel(X \mid E) = \frac{Bel(X)}{Bel(X) + Pl(E \setminus X)}.$$

Theorem 5.1 (Jaffray's theorem under mutual exclusivity). *As defined, the function $Bel(\cdot \mid E)$ is a belief function, where m' is its Möbius transform;*

$$m'(X) = \sum_{Y \subseteq X} (-1)^{|X \setminus Y|} Bel(Y \mid E), \quad \forall X \in 2^\Omega.$$

This function m' is then a valid basic belief assignment, alternatively expressed by

$$m'(X) = \sum_{T \in \mathcal{T}(X)} \tilde{m}(T).$$

Proof outline

As defined so far, m' is the Möbius transform of $Bel(\cdot \mid E)$ if and only if

$$Bel(X \mid E) = \sum_{Y \subseteq X} m'(Y) = \sum_{Y \subseteq X} \sum_{T \in \mathcal{T}(Y)} \tilde{m}(T), \quad \forall X \subseteq E.$$

Our aim is to establish this exact equivalence. By doing so, we confirm that m' is indeed the Möbius transform of $Bel(\cdot \mid E)$. Note that because m' is constructed entirely

from a summation of non-negative terms $\tilde{m}(T)$, proving this equivalence also shows that m' satisfies the non-negativity requirement of a basic belief assignment.

The proof consists of three parts. First, we find an alternative expression for the double summation over all sequences, based on a partial sum over $\tilde{m}$ which we denote by S . Next, we use backward induction to find an explicit expression that describes S . In the final step, we substitute this explicit formulation back into the summation and show its equivalence to the FH closed form.

Finding an alternate expression for the double summation

We aim to find a more tractable expression for the double summation

$$\sum_{Y \subseteq X} \sum_{T \in \mathcal{T}(Y)} \tilde{m}(T), \quad \forall X \subseteq E.$$

So we sum over $Y \subseteq X$ and $T \in \mathcal{T}(Y)$. Recognize now that any constructed set $Y = B \cup [(A_1 \cup A_2 \cup \dots \cup A_k) \cap E]$ will be a subset of X if and only if its distinct, mutually exclusive components are contained in X . Namely, $B \subseteq X$ and $(A_\ell \cap E) \subseteq X$ for all $1 \leq \ell \leq k$. Therefore, the double summation can be re-expressed as

$$\sum_{B \in \mathcal{B}(X)} \sum_{(A_1, A_2, \dots, A_k) \in \mathcal{S}(\emptyset)} \tilde{m}(B, A_1, A_2, \dots, A_k),$$

where $\mathcal{B}(X) = \{B \in \mathcal{B} : B \subseteq X\}$, and $\mathcal{S}(\emptyset)$ is the set of all finite sequences of distinct elements from $\mathcal{A}(X)$. Notice here the advantage of summing over ordered sequences rather than the unordered strings that Fagin and Halpern used. Namely, the fraction governing the weight of $\tilde{m}$ depends on the plausibility of the remaining space. Thus, the order in which the elements are removed changes the denominators, this then directly mirrors the sequential sensitivity of FH conditioning that was discussed in Section 5.2.

We next further decompose this sum by defining a notation for subsets of $\mathcal{S}(\emptyset)$ with a common initial sequence of length $1 \leq \ell \leq L$:

$$\mathcal{S}(A_1, A_2, \dots, A_\ell) = \{(\overline{A_1}, \overline{A_2}, \dots, \overline{A_k}) \in \mathcal{S}(\emptyset) : k \geq \ell \text{ and } \overline{A_j} = A_j, 1 \leq j \leq \ell\}.$$

We then denote the corresponding partial summation of weights for a given $B \in \mathcal{B}(X)$:

$$S(B, A_1, A_2, \dots, A_\ell) = \sum_{(\overline{A_1}, \overline{A_2}, \dots, \overline{A_k}) \in \mathcal{S}(A_1, A_2, \dots, A_\ell)} \tilde{m}(B, \overline{A_1}, \overline{A_2}, \dots, \overline{A_k}).$$

Under this new notation, $S(B)$ corresponds to $\mathcal{S}(\emptyset)$. Thus, our double summation is now simplified to $\sum_{B \in \mathcal{B}(X)} S(B)$. Hence, all that remains to be shown is that

$$Bel(X | E) = \sum_{B \in \mathcal{B}(X)} S(B).$$

Finding an explicit expression for S

We shall now derive an explicit expression for $S(B)$ by proceeding with a backward induction on the sequence length. Let $(A_1, A_2, \dots, A_L)$ be an ordering of all elements of $\mathcal{A}(X)$. The evidence set E_L is created such that all partially overlapping focal elements relevant to X have been removed. Hence, there is no ambiguity related to X in this setting; there exists no remaining partially overlapping focal element in $\mathcal{A}(X)$. This lack of ambiguity makes E_L a trivial setting, which is why it serves as the base case of our induction.

Fix $B \in \mathcal{B}(X)$. Every sequence of events in $\mathcal{S}(A_1, A_2, \dots, A_\ell)$, for $0 \leq \ell \leq L - 1$, except $(A_1, A_2, \dots, A_\ell)$, has at least an $(\ell + 1)$ -th term, A_j , where $\ell + 1 \leq j \leq L$. Consequently, we have the relation

$$S(B, A_1, \dots, A_\ell) = \tilde{m}(B, A_1, \dots, A_\ell) + \sum_{j=\ell+1}^L S(B, A_1, \dots, A_\ell, A_j), \quad \text{for } 0 \leq \ell \leq L - 1.$$

Now we pose our inductive hypothesis; for any $0 \leq \ell \leq L$, we have

$$S(B, A_1, A_2, \dots, A_\ell) = \tilde{m}(B, A_1, A_2, \dots, A_\ell) \times \frac{Pl(E_\ell)}{Pl(E_L)}.$$

For $\ell = L$, the set $\mathcal{S}(A_1, \dots, A_L)$ contains only the sequence $(A_1, \dots, A_L)$, so $S(B, A_1, \dots, A_L) = \tilde{m}(B, A_1, \dots, A_L)$.

Assume that the hypothesis holds for every ordered prefix of length $\ell + 1$. To shorten notations, let us define $T_\ell := (B, A_1, A_2, \dots, A_\ell)$ and $T_j := (B, A_1, \dots, A_\ell, A_j)$. We then

evaluate:

$$\begin{aligned} S(B, A_1, A_2, \dots, A_\ell) &= \tilde{m}(B, A_1, A_2, \dots, A_\ell) + \sum_{j=\ell+1}^L S(B, A_1, A_2, \dots, A_\ell, A_j) \\ &= \tilde{m}(T_\ell) + \sum_{j=\ell+1}^L S(T_j) \end{aligned}$$

which by inductive hypothesis gives

$$= \tilde{m}(T_\ell) + \sum_{j=\ell+1}^L \tilde{m}(T_j) \times \frac{Pl(E_\ell \setminus A_j)}{Pl(E_L)}$$

here we can substitute the recursive formula for $\tilde{m}(T_j)$ to get

$$\begin{aligned} &= \tilde{m}(T_\ell) + \sum_{j=\ell+1}^L \tilde{m}(T_\ell) \times \frac{m(A_j)}{Pl(E_\ell \setminus A_j)} \times \frac{Pl(E_\ell \setminus A_j)}{Pl(E_L)} \\ &= \tilde{m}(T_\ell) + \sum_{j=\ell+1}^L \tilde{m}(T_\ell) \times \frac{m(A_j)}{Pl(E_L)} \\ &= \frac{\tilde{m}(T_\ell)}{Pl(E_L)} \left[Pl(E_L) + \sum_{j=\ell+1}^L m(A_j) \right]. \end{aligned}$$

Finally, recall that we defined $E_0 = E, E_1 = E_0 \setminus A_1, E_2 = E_1 \setminus A_2, \dots$. Removing A_1 then exactly drops the plausibility by $m(A_1)$, such that $Pl(E_1) = Pl(E_0) - m(A_1)$. Since all sets in $\mathcal{A}$ are mutually exclusive, we can extend the same principle universally, so we see $Pl(E_\ell) = Pl(E_L) + \sum_{j=\ell+1}^L m(A_j)$. Substituting this equality yields

$$S(B, A_1, A_2, \dots, A_\ell) = \tilde{m}(T_\ell) \times \frac{Pl(E_\ell)}{Pl(E_L)}.$$

This shows that if our inductive hypothesis holds at order $(\ell + 1)$, then it does so too at order ℓ . Hence the explicit formulation for $S(B, A_1, A_2, \dots, A_\ell)$ holds.

In particular, for $\ell = 0$, we have the base sequence $T_0 = (B)$, which gives

$$S(B) = \tilde{m}(B) \times \frac{Pl(E_0)}{Pl(E_L)} = \frac{m(B)}{Pl(E)} \times \frac{Pl(E)}{Pl(E_L)} = \frac{m(B)}{Pl(E_L)}.$$

What then remains is to show that

$$Bel(X \mid E) = \sum_{B \in \mathcal{B}(X)} \frac{m(B)}{Pl(E_L)}.$$

Showing equivalence to the FH closed form

To complete the proof, we must find equivalence between

$$\frac{\sum_{B \in \mathcal{B}(X)} m(B)}{Pl(E_L)},$$

and the FH closed form:

$$\frac{Bel(X)}{Bel(X) + Pl(E \setminus X)}.$$

It is straightforward that the numerator $\sum_{B \in \mathcal{B}(X)} m(B)$ is equal to $Bel(X)$, since by definition $\mathcal{B}(X)$ contains all focal elements completely contained in X ;

$$\sum_{B \in \mathcal{B}(X)} m(B) = Bel(X).$$

For the denominator, let us evaluate the plausibility of E_L . Recall that $E_L = E \setminus (A_1 \cup A_2 \cup \dots \cup A_L)$. By definition, $Pl(E_L) = 1 - Bel(E_L^c) = 1 - Bel(E^c) - \sum_{j=1}^L m(A_j)$. Furthermore, the plausibility of the space outside X is $Pl(E \setminus X) = 1 - Bel(E^c \cup X) = 1 - Bel(E^c) - Bel(X) - \sum_{j=1}^L m(A_j)$.

Isolating the terms then shows $Pl(E_L) - Pl(E \setminus X) = Bel(X)$, which rearranges to $Pl(E_L) = Bel(X) + Pl(E \setminus X)$, giving us the correct denominator.

This demonstrates that m' is the Möbius transform of $Bel(\cdot \mid E)$. Because m' is built purely from nonnegative terms in the recurrence, its nonnegativity is self-evident. Therefore, m' is a valid basic belief assignment, and $Bel(\cdot \mid E)$ is a valid belief function. This is a reformulation of a restricted case of Jaffray's established theorem, rather than a separate result.

5.4. General adaptation

The adaptation of Jaffray's proof given in the previous section relies strictly on pairwise mutual exclusivity of all focal elements intersecting E . When focal elements A_j and A_k overlap, removing A_k from the conditioning event then removes $A_j \cap A_k$ as well. This simplification of the problem allowed us to view the recurrences of $\tilde{m}$ and the partial sum S as operations that change the set on which we are conditioning. The great algebraic advantage of this approach is that the mutual exclusivity gives us the equation

$$Pl(E_1) = Pl(E_0) - m(A_1).$$

Fortunately we can preserve the intuitive mechanics of this proof without the assumption of mutual exclusivity. Rather than sequentially restricting the conditioning event E , we keep E constant and sequentially transform the belief function itself. These transformed belief functions are auxiliary constructions and should not be interpreted as the results of successive applications of FH conditioning.

For a fixed set X , let $(A_1, \dots, A_L)$ be an ordering of the focal elements in $\mathcal{A}(X)$, so that the construction resolves the ambiguity related to X one focal element at a time. Let $Bel_0 = Bel$ be the initial belief function with corresponding bba $m_0 = m$, and plausibility function Pl_0 . We next define an iterative sequence of belief functions $Bel_1, Bel_2, \dots, Bel_L$. For each step $k \in \{1, \dots, L\}$, we construct the basic belief assignment m_k from m_{k-1} by shifting the mass of A_k entirely outside of E . In construction, we thus define a new set $C_k = A_k \setminus E$. The updated assignment m_k is then a duplicate of m_{k-1} for all subsets of Ω , except that we transfer the mass on A_k onto C_k , meaning that $m_k(A_k) = 0$, and $m_k(C_k) = m_{k-1}(C_k) + m_{k-1}(A_k)$.

We now evaluate the plausibility of E under this updated belief function. By definition, the plausibility is the sum of the masses of all focal elements that intersect E :

$$Pl_k(E) = \sum_{X \cap E \neq \emptyset} m_k(X).$$

Because C_k is defined by $C_k = A_k \setminus E$, it follows logically that $C_k \cap E = \emptyset$. Hence, the mass transferred to C_k contributes zero to $Pl_k(E)$. Furthermore, because in iteration k , the mass on A_k was reduced to 0, it also contributes zero to $Pl_k(E)$. Since all other focal elements remain the same from m_{k-1} , we see

$$Pl_k(E) = Pl_{k-1}(E) - m_{k-1}(A_k).$$

So instead of changing the set we condition on, we can change the underlying basic belief assignment iteratively. Through this method we can ensure a recursive drop in plausibility even when the sets in $\mathcal{A}$ overlap. And so we find an alternative recurrence for $\tilde{m}$ that holds without requiring mutual exclusivity

$$\tilde{m}(B, A_1, \dots, A_k) = \tilde{m}(B, A_1, \dots, A_{k-1}) \times \frac{Pl_{k-1}(E) - Pl_k(E)}{Pl_k(E)}.$$

For the fixed set X and ordering of $\mathcal{A}(X)$, we can use the same construction to reformulate Jaffray's partial sum S . Recall that $S(B, A_1, \dots, A_\ell)$ sums the values of $\tilde{m}$ over all sequences beginning with $(B, A_1, \dots, A_\ell)$. Every such sequence either ends at this point or continues with one of the remaining focal elements, hence

$$S(B, A_1, \dots, A_\ell) = \tilde{m}(B, A_1, \dots, A_\ell) + \sum_{j=\ell+1}^L S(B, A_1, \dots, A_\ell, A_j),$$

for $0 \leq \ell \leq L - 1$. At the final level all ambiguity related to X has been resolved, so that no A_i sets can be added to the sequence. Hence,

$$S(B, A_1, \dots, A_L) = \tilde{m}(B, A_1, \dots, A_L).$$

The backward induction then gives

$$S(B, A_1, \dots, A_\ell) = \tilde{m}(B, A_1, \dots, A_\ell) \frac{Pl_\ell(E)}{Pl_L(E)}.$$

For every ordering $(A_1, \dots, A_L)$, the auxiliary transformations reproduce exactly the denominators occurring in Jaffray's recurrence. Consequently, both the recurrence for $\tilde{m}$ and the explicit formulation of S remain valid without the assumption of mutual exclusivity. Summing the resulting sequence weights therefore produces the same conditioned bba m' as in Jaffray's original construction.

The full general proof resulting from this alternative construction is almost identical to the mutually exclusive version, hence it is given in Appendix B. This construction is therefore a reformulation of Jaffray's proof [3], intended to make the algebraic factors in the mass assignments interpretable as changes in plausibility.

5.5. Revisiting worked example

In Section 4.4, we worked out the conditional bba and corresponding belief in a set using Jaffray's auxiliary function $\tilde{m}$ and partial sum S . We now return to the same example to show how the general proof from Section 5.4 works in practice. We first construct the auxiliary belief functions, then show how the updated recursions and definitions produce the same results as in Section 4.4 based on Jaffray's original construction.

Recall from the worked example that $\Omega = \{x_1, \dots, x_7\}$, we wish to condition on $E = \{x_1, x_2, x_3, x_4, x_7\}$, we are interested in $Bel(X | E)$, where $X = \{x_1, x_2, x_3, x_7\}$, and the original belief function is defined by the bba values

$$\begin{array}{llllll} B_1 = \{x_1\} & B_2 = \{x_7\} & A_1 = \{x_2, x_5\} & A_2 = \{x_2, x_3, x_6\} & A_3 = \{x_4, x_5, x_6\} & C = \{x_5, x_6\} \\ m(B_1) = \frac{1}{16} & m(B_2) = \frac{1}{8} & m(A_1) = \frac{1}{4} & m(A_2) = \frac{3}{16} & m(A_3) = \frac{1}{8} & m(C) = \frac{1}{4}. \end{array}$$

As before, $\mathcal{B} = \{B_1, B_2\}$ contains the focal elements enclosed in E , while $\mathcal{A} = \{A_1, A_2, A_3\}$ contains the partially overlapping focal elements. The set C is the only focal set completely outside of E , hence $Pl(E) = 1 - m(C) = \frac{3}{4}$. Note furthermore that

$$A_1 \cap E = \{x_2\} \subseteq A_2 \cap E = \{x_2, x_3\}.$$

Therefore, the focal elements in $\mathcal{A}$ are not mutually exclusive, so the mutually exclusive proof from Section 5.3 cannot be applied. So instead, we use the general adaptation from Section 5.4.

Now we can construct the auxiliary belief functions described in the general proof. Recall from our construction that we begin with $m_0 = m$ and that at each new iteration we move the mass from a partially overlapping set A_i to outside of E , so that the terminal auxiliary bba contains no partially overlapping focal elements. At step k , we define $C_k = A_k \setminus E$ and construct m_k by transferring the complete mass on A_k onto C_k . Hence,

$$m_k(A_k) = 0, \quad m_k(C_k) = m_{k-1}(C_k) + m_{k-1}(A_k),$$

while all other masses remain unchanged. Since C_k lies outside of E , this gives the recursion on Pl_k

$$Pl_k(E) = Pl_{k-1}(E) - m_{k-1}(A_k).$$

The point to stress here is that these transformed belief functions are auxiliary. So their corresponding bbas are not obtained through successive applications of FH conditioning and should not be confused with the final assignment m' .

Turning now to our example, we have three elements in $\mathcal{A}$. Let us resolve all three elements of $\mathcal{A}$ in the order (A_1, A_2, A_3) to illustrate the complete auxiliary transformation of the original bba. In the first iteration, we have $C_1 = A_1 \setminus E = \{x_5\}$. We therefore transfer the mass $\frac{1}{4}$ from A_1 onto $\{x_5\}$. Since this mass no longer intersects E , the plausibility decreases from $Pl_0(E) = \frac{3}{4}$ to

$$Pl_1(E) = Pl_0(E) - m_0(A_1) = \frac{3}{4} - \frac{1}{4} = \frac{1}{2}.$$

The remaining iterations follow identically and are given below.

k	resolved set	mass moved outside of E	$Pl_k(E)$
0	—	—	$\frac{3}{4}$
1	A_1	$m_0(A_1) = \frac{1}{4}$	$\frac{1}{2}$
2	A_2	$m_1(A_2) = \frac{3}{16}$	$\frac{5}{16}$
3	A_3	$m_2(A_3) = \frac{1}{8}$	$\frac{3}{16}$

Look here specifically at the last row. The belief function represented there has no partially overlapping focal elements, so every focal element is either contained in E or in $\Omega \setminus E$. So the masses of all three partially overlapping focal elements have been transferred outside of E . Therefore, only B_1 and B_2 still contribute to the plausibility of E , giving

$$Pl_3(E) = m(B_1) + m(B_2) = \frac{1}{16} + \frac{1}{8} = \frac{3}{16}.$$

Recall then that specifically the partially overlapping focal elements caused difficulty in evaluating the conditional belief. So the belief function Bel_3 is constructed specifically for E , so that evaluating $Bel_3(Y | E)$ is now a trivial task. In essence, we have resolved all partially overlapping mass, resulting in a simple base case to build upon.

We now use these transformed plausibilities to reconstruct Jaffray's auxiliary function. Recall from Section 5.4 the recursive definition $\tilde{m}(B) = \frac{m(B)}{Pl_0(E)}$, and

$$\tilde{m}(B, A_1, \dots, A_k) = \tilde{m}(B, A_1, \dots, A_{k-1}) \frac{Pl_{k-1}(E) - Pl_k(E)}{Pl_k(E)}.$$

The numerator here is the mass that has just been transferred outside of E , while the denominator is the plausibility that remains after this transfer. So let us now reperform the calculations from the worked example in Section 4.4. For the base sequence beginning with B_1 , we again find

$$\tilde{m}(B_1) = \frac{m(B_1)}{Pl_0(E)} = \frac{1/16}{3/4} = \frac{1}{12}.$$

In the first iteration, after resolving A_1 , the recursive factor equals

$$\frac{Pl_0(E) - Pl_1(E)}{Pl_1(E)} = \frac{3/4 - 1/2}{1/2} = \frac{1}{2},$$

and therefore $\tilde{m}(B_1, A_1) = 1/12 \times 1/2 = 1/24$. Resolving A_2 next gives us

$$\tilde{m}(B_1, A_1, A_2) = \frac{1}{24} \frac{Pl_1(E) - Pl_2(E)}{Pl_2(E)} = \frac{1}{24} \frac{1/2 - 5/16}{5/16} = \frac{1}{40}$$

The reverse order uses a different sequence of auxiliary belief functions. Resolving A_2 first reduces the plausibility from $\frac{3}{4}$ to $\frac{9}{16}$, after which resolving A_1 reduces it to $\frac{5}{16}$. Hence,

$$\tilde{m}(B_1, A_2, A_1) = \frac{1}{12} \frac{3/4 - 9/16}{9/16} \frac{9/16 - 5/16}{5/16} = \frac{1}{45}.$$

Both orders end with the same remaining plausibility $\frac{5}{16}$, but the different intermediate plausibilities produce different recursive factors and therefore different values of $\tilde{m}$. The auxiliary belief functions therefore make visible how the intermediate plausibilities produce the factors already present in Jaffray's recurrence. The results found are exactly the same as those calculated directly from Jaffray's recurrence in Section 4.4. The construction through auxiliary belief functions therefore provides an interpretation for the factors already present in $\tilde{m}$. Since the same values of $\tilde{m}$ are obtained, summing them over sequences that construct the same set produces the same conditioned bba m' as in Section 4.4.

Let us continue now by reverifying the value $Bel(X | E) = 3/5$ for $X = \{x_1, x_2, x_3, x_7\}$ as we saw in Section 4.4. Recall that $\mathcal{B}(X) = \{B_1, B_2\}$, $\mathcal{A}(X) = \{A_1, A_2\}$. Hence, for the calculation of $Bel(X | E)$ the terminal level is $L = 2$: resolving all ambiguity related to X leaves A_3 unchanged. After transferring the masses on A_1 and A_2 outside of E , the remaining plausibility relevant to this partial sum is

$$Pl_2(E) = Pl(E) - m(A_1) - m(A_2) = \frac{3}{4} - \frac{1}{4} - \frac{3}{16} = \frac{5}{16}.$$

Recall that the recursion for $S(T)$ sums all values of $\tilde{m}$ over sequences that start with sequence T . Recall then the explicit formulation for S from Section 5.4;

$$S(B, A_1, \dots, A_\ell) = \tilde{m}(B, A_1, \dots, A_\ell) \frac{Pl_\ell(E)}{Pl_L(E)}.$$

Hence, we see

$$S(B_1) = \frac{1/16}{5/16} = \frac{1}{5}, \quad S(B_2) = \frac{1/8}{5/16} = \frac{2}{5},$$

and therefore

$$Bel(X | E) = \sum_{B \in \mathcal{B}(X)} S(B) = \frac{3}{5}.$$

This example makes our adapted general proof concrete. Each transformation removes exactly one source of ambiguity from the plausibility of E , even when mutual exclusivity does not hold. The resulting changes in plausibility produce the recursive factors in $\tilde{m}$. Although different transformation orders may receive different weights, summing all orders that construct the same set produces the corresponding order-agnostic bba m' . Finally, summing the sequence weights through S produces the conditional belief.

6. Interpretation and conclusion

6.1. The epistemic role of noncommutativity

Recall from Section 5.1 that successive FH conditioning is generally noncommutative. So in general, for events $X, E_1, E_2 \subseteq \Omega$ we have that $Bel((X | E_1) | E_2)$ need not equal $Bel((X | E_2) | E_1)$. Both Fagin and Halpern [2] and Jaffray [3] discuss the noncommutative property in their respective papers. One reason this property has received attention is that Dempster's conditioning is commutative, while the FH definition was motivated partly by difficulties associated with Dempster's conditioning. Some scholars have viewed this noncommutative property as a weakness of FH conditioning. For instance, Pearl stated the following when comparing shortcomings of Dempster's conditioning to the qualities of the conditioning rule by Fagin and Halpern: "Unfortunately, although FH conditioning alleviates some of the problems encountered with belief functions, it creates new ones. First, FH conditioning is not commutative, and hence we cannot process multiple evidence sequentially upon arrival" [5, p. 381].

An epistemic perspective underlying such a critique is that conditioning should represent the accumulation of facts in random order about one fixed outcome. In such a perspective, the evidence consists of facts about one outcome and learning one event does not affect which later evidence will be found. This perspective is natural in most settings in classical probability theory. Consider for example revealing first the face, then the color of a playing card. Having received this evidence in the opposite order would not have changed the relevant information. All that is relevant is that the card lies in the intersection of both events.

One can imagine settings in which receiving new information changes the process of accumulation of new information. Consider a forensic investigation in which some laboratory evidence E_1 is received. Based on this, it is decided which locations to further examine, through which new evidence E_2 is found. The investigation in which E_2 was found was directed first by finding E_1 . Therefore, the explanatory value of E_2 should be viewed relative to what E_1 already implied. In that sense, reversing the order of incoming evidence can impact the epistemic outcome. Say we modelled this setting with belief functions. Reversing the order of conditioning then generally results in a different intermediate belief function. Since the second conditioning operation acts on this intermediate belief function, ambiguous masses are then resolved according to different plausibility structures. This does not show that conditional belief formally models such a dynamic process correctly. It does however provide an explanation of when noncommutativity may be epistemically useful. If earlier information affects the process leading to new information, then forgetting the order also forgets relevant parts

of the process. Noncommutativity then represents a relevant difference between two histories rather than introducing an arbitrary dependence on the order in which facts are revealed.

Conditional belief can nevertheless still be applied in settings where order should not matter. When E_1 and E_2 are facts revealed in random order and no relevant action occurs between them, the evidence can be combined into the event $E_1 \cap E_2$. Provided $Bel(E_1 \cap E_2) > 0$, one then conditions the original belief function directly on this intersection using $Bel(\cdot \mid E_1 \cap E_2)$. The result is then independent of the order in which the two facts were revealed.

To conclude, noncommutativity does not have to be a flaw of conditional belief. Its suitability depends on what the conditioning operation is intended to perform. When finding one piece of evidence does not affect which later evidence is found, one should condition directly on the intersection of the events. When earlier evidence changes the decisions, observations, or context for later evidence, then the order of conditioning can carry relevant epistemic information.

6.2. The sequence-based structure of conditional mass

Recall Jaffray's auxiliary function $\tilde{m}$ and the resulting construction of m' from Chapter 4 [3]. The interpretation of the recursive definition of $\tilde{m}$ through successive changes in plausibility is developed in Chapter 5 of this thesis. Throughout this thesis, the basic belief assignment has been presented as the lowest level representation of a belief function. The conditioned bba m' similarly assigns the final conditional mass directly to subsets of the outcome space. However, Jaffray's construction may indicate that there is a more detailed level beneath m' . Namely, the value $m'(X)$ is obtained by summing the values of $\tilde{m}$ over all ordered sequences T that construct the set X . Different sequences may construct the same set while having different values of $\tilde{m}$. Once $m'(X)$ is constructed from these values, the individual orderings represented by the sequences can no longer be distinguished. Thus, m' assigns masses to sets, while $\tilde{m}$ retains information about the order of ambiguity resolution.

Chapter 5 provided us with an interpretation for these ordered sequences. Each additional focal element in a sequence produces a reduction in the plausibility remaining after the preceding focal elements have been accounted for. The next value of $\tilde{m}$ is obtained by multiplying the previous value by a factor determined by this change in plausibility. Under mutual exclusivity this factor can be expressed as a relative increase in conditional belief. In the general case, the same changes were reproduced through auxiliary transformations of the bba. Under this interpretation, a sequence $T = (B, A_1, \dots, A_k)$ records one order in which the ambiguities contained in the focal elements $A_1, \dots, A_k$ are resolved relative to the focal element B .

There is also a mathematical indication that this sequential perspective may be more direct than the final conditioned bba. Namely, Jaffray uses his partial sum S to calculate

conditional belief by summing the relevant values of $\tilde{m}$. This circumvents the need to calculate any value of m' to be able to calculate conditional belief. Formally, for every $X \subseteq E$, the proof reaches the characterization

$$Bel(X | E) = \sum_{Y \subseteq X} \sum_{T \in \mathcal{T}(Y)} \tilde{m}(T).$$

The construction of m' is necessary in the proof only to show that the resulting function has a standard bba, but it is not necessary for calculating the conditional belief itself.

This suggests a possible direction for further research. Namely, the question is whether assignments on ordered sequences could provide a foundational representation of belief more generally, thus providing an alternative to the bba as the most fundamental layer in the theory. This thesis does not establish such a representation. Jaffray's $\tilde{m}$ is still derived from an original bba m and a conditioning event E . Further research would have to determine which mathematical conditions an arbitrary assignment on ordered sequences should satisfy to form a valid foundational representation of belief. Whether every belief function can have such a representation, and how an ordinary bba can be recovered from it are also unknowns. If such an equivalent representation can be established, the central theorem may admit a simpler proof, since Jaffray's construction could then be verified directly at the level of sequence weights, circumventing the construction of m' .

Further research could also look for a more precise epistemic or probabilistic interpretation of $\tilde{m}$. The interpretation developed in this thesis describes $\tilde{m}(B, A_1, \dots, A_k)$ as the weight assigned to one path of ambiguity resolution, where B gives the initial unambiguous support and the ordered focal elements $A_1, \dots, A_k$ record successive reductions in the remaining plausibility. This explains the mathematical role of the ordered sequences, but it does not yet identify an epistemic process or probabilistic experiment that would result in the values of $\tilde{m}$. We have compared this internal ordering to sequential conditioning, however we could not determine whether these processes reflect the same underlying epistemic process. A more complete interpretation would therefore have to explain what the sequence weights represent, and under which epistemic or probabilistic process these weights could be found.

6.3. Conclusion and limitations

This thesis investigated how and why FH conditional belief is itself a belief function. The proof by Fagin and Halpern reconstructs the conditioned bba through an inclusion-exclusion mechanism, which makes the nonnegativity of the resulting bba difficult to see directly. Jaffray instead constructs the conditioned bba by summing nonnegative weights assigned to ordered sequences. Comparing these proofs showed that the recursive factors in Jaffray's auxiliary function $\tilde{m}$ can be interpreted through successive reductions in plausibility. This interpretation led first to an adaptation under the assumption of mutual exclusivity, and then to a general adaptation based on auxiliary transformations

of the bba. Together, these results explain where the sequential structure enters Jaffray's construction, and how the construction preserves the nonnegativity required of the conditioned bba.

The thesis remains limited to finite outcome spaces. Extending FH conditioning to infinite outcome spaces would therefore be an interesting direction for further research. The thesis also identifies two distinct forms of sensitivity to order that must be compared with caution: the order of events in successive FH conditioning, and the order of focal elements in Jaffray's auxiliary function $\tilde{m}$. Although both operations can be related to reductions in plausibility, the interpretation of $\tilde{m}$ relies on auxiliary mathematical transformations whose correspondence to an epistemic or empirical updating process has not been established. Furthermore, $\tilde{m}$ is still derived from a bba m that is agnostic to order, so this thesis does not establish a sequence based representation of belief. Whether such a representation exists, and whether $\tilde{m}$ admits a precise epistemic or probabilistic interpretation, remain topics for further research.

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A. Fagin and Halpern continued

This appendix contains the remainder of the proof by Fagin and Halpern [2]. What remains is to prove requirement B ($m'(X) \geq 0 \quad \forall X \subseteq E$).

Recall that m' is a sum of m'' terms, hence it suffices to show nonnegativity of m'' . Note that the current definition of m'' is recursive, recall furthermore from Example 3.4 that through the inclusion exclusion mechanism, a lot of terms cancel out when evaluating m'' , leaving a neat final construction. The way FH prove B is by first proving an explicit formula for m'' which will look familiar to the results from our Example 3.4. Then the nonnegativity of this explicit m'' will be proven through calculus. So what we must show is that $m''(B_i A'_{j_1} \dots A'_{j_r})$ is nonnegative. For ease of notation we replace i by 1, and $j_1, \dots, j_r$ by $1, \dots, r$, so we show $m''(B_1 A'_1 \dots A'_r)$ to be nonnegative. The explicit formula to show is then

$$m''(B_1 A'_1 \dots A'_r) = \sum_{I \subseteq \{1, \dots, r\}} (-1)^{r-|I|} \frac{b_1}{1 - \sum_{j \in I} a_j}. \quad (\text{A.1})$$

We prove nonnegativity of (A.1) by induction on r . The base case, $r = 0$, holds as we have $m''(B_1) = b_1 > 0$. For the induction step, we assume that the explicit formula (A.1) holds for all proper substrings Y of $A'_1 \dots A'_r$. We can then substitute this formula into the right-hand side of our recursive definition (3.1) for every $m''(B_1 Y)$. To keep the notations manageable, let us denote the recurring fraction $\frac{b_1}{1 - \sum_{j \in I} a_j}$ by E_I . Our goal is to compute the coefficient of each E_I in the resulting substituted expression and show it matches the coefficient in our explicit formula. First, consider the case where I is the full set $\{1, \dots, r\}$. It is clear from our substitution into (3.1) that E_I appears only once, as the leading positive term. In our explicit formula (A.1), the coefficient for this full set is $(-1)^{r-r} = 1$, which matches the required coefficient.

Now let us consider a subset I with size $|I| = s$, and assume $s < r$. We must carefully count how many subsets of $\{1, \dots, r\}$ contain I . The number of such subsets of size t is 0 if $t < s$, and exactly $\binom{r-s}{t-s}$ otherwise. Because of our induction hypothesis, each of these proper substrings Y of length t contributes an E_I term with a coefficient of $(-1)^{t-s}$. However, recall that in our recursive definition (3.1), these $m''(B_1 Y)$ terms are being subtracted. Hence, the total contribution of E_I from all proper substrings of length t becomes $(-1)^{t-s+1} \binom{r-s}{t-s} E_I$. Summing this over all valid lengths t , the total contribution of E_I is

$$E_I \sum_{t=s}^{r-1} (-1)^{t-s+1} \binom{r-s}{t-s}.$$

Note that the largest index of the sum stops at $t = r - 1$ rather than $t = r$, because our recursive formula (3.1) only subtracts proper substrings. We can simplify this sum by

changing the index. Let $u = t - s$. The expression then becomes

$$E_I \sum_{u=0}^{r-s-1} (-1)^{u+1} \binom{r-s}{u}. \quad (\text{A.2})$$

To evaluate this, we use the binomial theorem. We know that the full alternating sum of these binomial coefficients equals zero:

$$\sum_{u=0}^{r-s} (-1)^{u+1} \binom{r-s}{u} = 0,$$

because it is simply -1 times the binomial expansion of $(1-1)^{r-s}$. Therefore, our sum in (A.2), which is missing only the final $u = r-s$ term, must evaluate to the negative of that missing term. That missing term is $(-1)^{r-s+1} \binom{r-s}{r-s} E_I = (-1)^{r-s+1} E_I$. Taking the negative of this gives us $(-1)^{r-s} E_I$. Since $s = |I|$, this means the coefficient of E_I is precisely $(-1)^{r-|I|}$. This is exactly the coefficient required by our explicit formula (A.1), completing the induction.

To prove that m'' is nonnegative, we now need only to show nonnegativity of our explicit formula. This, after dividing by the positive b_1 resolves to showing

$$\sum_{I \subseteq \{1, \dots, r\}} (-1)^{r-|I|} \frac{1}{1 - \sum_{j \in I} a_j} \geq 0.$$

Note here that each a_j is positive and that $\sum_{j=1}^r a_j < 1$. Indeed, the normalized masses satisfy

$$\sum_{i=1}^n a_i + \sum_{i=1}^t b_i = 1,$$

while $Bel(E) > 0$ implies that at least one b_i is positive. Hence, for every $I \subseteq \{1, \dots, r\}$ we have $0 \leq \sum_{j \in I} a_j < 1$, and so we may use the geometric series

$$\sum_{k=0}^{\infty} \left(\sum_{j \in I} a_j \right)^k = \frac{1}{1 - \sum_{j \in I} a_j}.$$

Here we interpret $\left(\sum_{j \in I} a_j \right)^0 = 1$ even when $\sum_{j \in I} a_j = 0$. So by now we have reduced the task of proving nonnegativity of m'' to showing that for each r and each k we have

$$\sum_{I \subseteq \{1, \dots, r\}} (-1)^{r-|I|} \left(\sum_{j \in I} a_j \right)^k \geq 0.$$

Define $f_{r,k}$ to be a function with r arguments, with domain those tuples $(a_1, \dots, a_r)$ where each a_i is nonnegative, and

$$f_{r,k}(a_1, \dots, a_r) = \sum_{I \subseteq \{1, \dots, r\}} (-1)^{r-|I|} \left(\sum_{j \in I} a_j \right)^k.$$

Here we make the convention that $f_{0,0} = 1$, while $f_{0,k} = 0$ for every $k > 0$. This agrees with our convention that the empty sum raised to the zeroth power equals 1. To prove our nonnegativity condition, we need only show that each $f_{r,k}$ is nonnegative. To assist in this, define an auxiliary function $g_{r,k}$ with the same domain as $f_{r,k}$, such that

$$g_{r,k}(a_1, \dots, a_r) = \sum_{I \subseteq \{1, \dots, r\}, r \in I} (-1)^{r-|I|} \left(\sum_{j \in I} a_j \right)^k$$

if $r > 0$, and let $g_{0,k} = 0$ for every k . So, $f_{r,k}$ sums over all subsets of $\{1, \dots, r\}$, while $g_{r,k}$ sums only over those subsets that contain r . We shall show, by induction on r , that for every k , the functions $f_{r,k}$ and $g_{r,k}$ are nonnegative. The base case ($r = 0$) is immediate from our convention. Assume inductively that $r \geq 1$, and that $f_{r-1,k}$ and $g_{r-1,k}$ are nonnegative for every k . We now show, by a second induction on k , that $f_{r,k}$ and $g_{r,k}$ are nonnegative. For the base case ($k = 0$), note that

$$f_{r,0}(a_1, \dots, a_r) = \sum_{I \subseteq \{1, \dots, r\}} (-1)^{r-|I|} = 0,$$

since it is the same as the binomial expansion of $(1 - 1)^r$. Now for $g_{r,0}$, we have

$$g_{r,0}(a_1, \dots, a_r) = \sum_{I \subseteq \{1, \dots, r\}, r \in I} (-1)^{r-|I|} = \sum_{I' \subseteq \{1, \dots, r-1\}} (-1)^{r-1-|I'|} = f_{r-1,0}(a_1, \dots, a_{r-1}),$$

so $g_{r,0}$ is also nonnegative.

For the inductive step on k , assume that $f_{r,k-1}$ and $g_{r,k-1}$ are nonnegative. To show that $f_{r,k}$ and $g_{r,k}$ are nonnegative, we shall show that $f_{r,k}(a_1, \dots, a_r)$ and $g_{r,k}(a_1, \dots, a_r)$ are nonnegative when $a_r = 0$, and that their first derivatives with respect to a_r are nonnegative. If $a_r = 0$, then we can split the sum for $f_{r,k}$ over the subsets that contain r and those that do not:

$$\begin{aligned} f_{r,k}(a_1, \dots, a_r) &= \sum_{I \subseteq \{1, \dots, r\}} (-1)^{r-|I|} \left(\sum_{j \in I} a_j \right)^k \\ &= \sum_{I \subseteq \{1, \dots, r\}, r \in I} (-1)^{r-|I|} \left(\sum_{j \in I} a_j \right)^k + \sum_{I \subseteq \{1, \dots, r\}, r \notin I} (-1)^{r-|I|} \left(\sum_{j \in I} a_j \right)^k \\ &= \sum_{I' \subseteq \{1, \dots, r-1\}} (-1)^{r-1-|I'|} \left(\sum_{j \in I'} a_j \right)^k + \sum_{I \subseteq \{1, \dots, r-1\}} (-1)^{r-|I|} \left(\sum_{j \in I} a_j \right)^k \\ &= f_{r-1,k}(a_1, \dots, a_{r-1}) - f_{r-1,k}(a_1, \dots, a_{r-1}) \\ &= 0. \end{aligned}$$

If $a_r = 0$, then

$$g_{r,k}(a_1, \dots, a_{r-1}, 0) = f_{r-1,k}(a_1, \dots, a_{r-1}),$$

which is nonnegative by our inductive assumption on r . As for the derivatives: the derivative of $f_{r,k}(a_1, \dots, a_r)$ with respect to a_r is

$$k \sum_{I \subseteq \{1, \dots, r\}, r \in I} (-1)^{r-|I|} \left(\sum_{j \in I} a_j \right)^{k-1} = kg_{r,k-1}(a_1, \dots, a_r),$$

which is nonnegative by our inductive assumption on k . The derivative of $g_{r,k}(a_1, \dots, a_r)$ with respect to a_r is exactly $kg_{r,k-1}(a_1, \dots, a_r)$, which again is nonnegative by the inductive assumption on k . Since both functions evaluate to nonnegative values at $a_r = 0$ and have nonnegative rates of change with respect to a_r , the functions must be nonnegative for all $a_r \geq 0$.

This then concludes the proof of B. Having satisfied all four conditions (A, B, C, and D), we have formally established that m' is a valid basic belief assignment. This proves that Bel' is a belief function, which in turn proves that FH conditional belief $Bel(\cdot \mid E)$ is a valid belief function.

B. General adaptation

In this appendix, we present an adapted version of Jaffray's proof [3] developed in Section 5.4, that does not require mutual exclusivity of the focal elements. In Jaffray's proof there are often individual basic belief assignments or sums thereof that are not easily interpretable. This version replaces many references to the individual weights assigned to sets by formulations in belief or plausibility. Here we preserve the intuitive mechanics of the proof in Section 5.3, while proving Jaffray's theorem in its general form.

Notation and setup

Let Ω be a finite outcome space, let Bel be some belief function on 2^Ω , and let m be the corresponding basic belief assignment. Let then $E \subseteq \Omega$ be the evidence set we wish to condition on, where we assume $Bel(E) > 0$. Let $\mathcal{B}$ denote the set of focal elements of Bel that are enclosed in E , and let $\mathcal{A}$ denote the set of focal elements of Bel that intersect both E and $\Omega \setminus E$. No assumption of mutual exclusivity is imposed on the focal elements in $\mathcal{A} \cup \mathcal{B}$.

Take $\mathcal{L}$ to be the set of all probability measures on $(\Omega, 2^\Omega)$, and let $\mathcal{P}$ be the collection of probability measures consistent with Bel ; $\mathcal{P} = \{P \in \mathcal{L} : P \geq Bel\}$. For a fixed target set $X \subseteq E$, define $\mathcal{A}(X) = \{A \in \mathcal{A} : A \cap E \subseteq X\}$. Throughout the remainder of this chapter, resolving the ambiguity related to X means removing from $Pl_0(E)$ the contributions of exactly the focal elements in $\mathcal{A}(X)$. Focal elements in $\mathcal{A} \setminus \mathcal{A}(X)$ may still partially overlap E , but they intersect $E \setminus X$ and therefore remain part of the competing plausibility $Pl_0(E \setminus X)$. Hence, after removing all ambiguity related to X , there can still be other partially overlapping focal elements in the original belief function. We next define an iterative sequence of belief functions $Bel_0, Bel_1, \dots, Bel_L$, where $m_0 = m$, and for each step k , the bba m_k is obtained from m_{k-1} by transferring the mass of A_k entirely outside of E onto the set $C_k = A_k \setminus E$. More precisely, we define $m_k(A_k) = 0$, $m_k(C_k) = m_{k-1}(C_k) + m_{k-1}(A_k)$, and leave all other masses unchanged. This yields a corresponding sequence of plausibility functions $Pl_0, Pl_1, \dots, Pl_L$. These belief functions are auxiliary constructions and do not represent successive applications of FH conditioning. Conceptually, applying this construction to the elements of $\mathcal{A}(X)$ removes the ambiguity related to X one focal element at a time, producing a predictable reduction in plausibility at every step. We can then add these focal elements back one by one, where each addition restores a predictable amount of plausibility.

For an ordered sequence of focal elements $T = (B, A_1, A_2, \dots, A_k)$ where $B \in \mathcal{B}$, and the $A_i \in \mathcal{A}$ are all distinct, we define the recursive auxiliary function $\tilde{m}(T)$ by

$$\tilde{m}(B) = \frac{m(B)}{Pl_0(E)}, \text{ and}$$

$$\tilde{m}(B, A_1, A_2, \dots, A_k) = \tilde{m}(B, A_1, A_2, \dots, A_{k-1}) \times \frac{Pl_{k-1}(E) - Pl_k(E)}{Pl_k(E)}.$$

Let $\mathcal{T}(X)$ for $X \subseteq \Omega$ denote the set of all ordered sequences $T = (B, A_1, A_2, \dots, A_k)$ that construct the set X :

$$\mathcal{T}(X) = \left\{ T = (B, A_1, \dots, A_k) \mid B \cup \left(\bigcup_{i=1}^k A_i \cap E \right) = X \right\}.$$

Finally, define $\mathcal{P}^E = \{P(\cdot \mid E) \text{ for } P \in \mathcal{P}\}$, so that FH conditional belief admits the conditioned lower-envelope representation

$$Bel(X \mid E) = \min\{Q(X) : Q \in \mathcal{P}^E\}.$$

As established before, for $X \subseteq E$ this is equivalent to the FH closed form

$$Bel(X \mid E) = \frac{Bel(X)}{Bel(X) + Pl_0(E \setminus X)}.$$

Theorem B.1. *As defined, the function $Bel(\cdot \mid E)$ is a belief function, where m' is its Möbius transform;*

$$m'(X) = \sum_{Y \subseteq X} (-1)^{|X \setminus Y|} Bel(Y \mid E), \quad \forall X \in 2^\Omega.$$

This function m' is then a valid basic belief assignment, alternatively expressed by

$$m'(X) = \sum_{T \in \mathcal{T}(X)} \tilde{m}(T).$$

Proof outline

As defined so far, m' is the Möbius transform of $Bel(\cdot \mid E)$ if and only if

$$Bel(X \mid E) = \sum_{Y \subseteq X} m'(Y) = \sum_{Y \subseteq X} \sum_{T \in \mathcal{T}(Y)} \tilde{m}(T), \quad \forall X \subseteq E.$$

Our aim is to establish this exact equivalence. By doing so, we confirm that m' is indeed the Möbius transform of $Bel(\cdot \mid E)$. Note that because m' is constructed entirely from a summation of non-negative terms $\tilde{m}(T)$, proving this equivalence also shows that m' satisfies the non-negativity requirement of a basic belief assignment.

The proof consists of three parts. First, we find an alternative expression for the double summation over all sequences, based on a partial sum over $\tilde{m}$ which we denote by S . Next, we use backward induction to find an explicit expression that describes S . In the final step, we substitute this explicit formulation back into the summation and show its equivalence to the FH closed form.

Finding an alternate expression for the double summation

We aim to find a more tractable expression for the double summation

$$\sum_{Y \subseteq X} \sum_{T \in \mathcal{T}(Y)} \tilde{m}(T), \quad \forall X \subseteq E.$$

So we sum over $Y \subseteq X$ and $T \in \mathcal{T}(Y)$. Recognize now that any constructed set $Y = B \cup [(A_1 \cup A_2 \cup \dots \cup A_k) \cap E]$ will be a subset of X if and only if its distinct components are contained in X . Namely, $B \subseteq X$ and $(A_\ell \cap E) \subseteq X$ for all $1 \leq \ell \leq k$. Therefore, the double summation can be re-expressed as

$$\sum_{B \in \mathcal{B}(X)} \sum_{(A_1, A_2, \dots, A_k) \in \mathcal{S}(\emptyset)} \tilde{m}(B, A_1, A_2, \dots, A_k),$$

where $\mathcal{B}(X) = \{B \in \mathcal{B} : B \subseteq X\}$, and $\mathcal{S}(\emptyset)$ is the set of all finite sequences of distinct elements from $\mathcal{A}(X)$. Notice here the advantage of summing over ordered sequences rather than the unordered strings that Fagin and Halpern used. Namely, the fraction governing the weight of $\tilde{m}$ depends on the plausibility of the remaining space. Thus, the order in which the elements are removed changes the denominators, this then directly mirrors the sequential sensitivity of FH conditioning that was discussed in Section 5.2.

We next further decompose this sum by defining a notation for subsets of $\mathcal{S}(\emptyset)$ with a common initial sequence of length $1 \leq \ell \leq L$:

$$\mathcal{S}(A_1, A_2, \dots, A_\ell) = \{(\overline{A_1}, \overline{A_2}, \dots, \overline{A_k}) \in \mathcal{S}(\emptyset) : k \geq \ell \text{ and } \overline{A_j} = A_j, 1 \leq j \leq \ell\}.$$

We then denote the corresponding partial summation of weights for a given $B \in \mathcal{B}(X)$:

$$S(B, A_1, A_2, \dots, A_\ell) = \sum_{(\overline{A_1}, \overline{A_2}, \dots, \overline{A_k}) \in \mathcal{S}(A_1, A_2, \dots, A_\ell)} \tilde{m}(B, \overline{A_1}, \overline{A_2}, \dots, \overline{A_k}).$$

Under this new notation, $S(B)$ corresponds to $\mathcal{S}(\emptyset)$. Thus, our double summation is now simplified to $\sum_{B \in \mathcal{B}(X)} S(B)$. Hence, all that remains to be shown is that

$$Bel(X | E) = \sum_{B \in \mathcal{B}(X)} S(B).$$

Finding an explicit expression for S

We shall now derive an explicit expression for $S(B)$ by proceeding with a backward induction on the sequence length. Let $(A_1, A_2, \dots, A_L)$ be an ordering of all elements of $\mathcal{A}(X)$. The auxiliary belief function Bel_L is created such that all partially overlapping focal elements relevant to X have been shifted outside of E . Hence, there is no ambiguity related to X in this setting; there exists no remaining partially overlapping focal element in $\mathcal{A}(X)$. This lack of ambiguity makes Bel_L a trivial setting, which is why it serves as the base case of our induction.

Fix $B \in \mathcal{B}(X)$. Every sequence of events in $\mathcal{S}(A_1, A_2, \dots, A_\ell)$, for $0 \leq \ell \leq L-1$, except $(A_1, A_2, \dots, A_\ell)$, has at least an $(\ell+1)$ -th term, A_j , where $\ell+1 \leq j \leq L$. Consequently, we have the relation

$$S(B, A_1, \dots, A_\ell) = \tilde{m}(B, A_1, \dots, A_\ell) + \sum_{j=\ell+1}^L S(B, A_1, \dots, A_\ell, A_j), \quad \text{for } 0 \leq \ell \leq L-1.$$

Now we pose our inductive hypothesis; for any $0 \leq \ell \leq L$, we have

$$S(B, A_1, A_2, \dots, A_\ell) = \tilde{m}(B, A_1, A_2, \dots, A_\ell) \times \frac{Pl_\ell(E)}{Pl_L(E)}.$$

For $\ell = L$, the set $\mathcal{S}(A_1, \dots, A_L)$ contains only the sequence $(A_1, \dots, A_L)$, so $S(B, A_1, \dots, A_L) = \tilde{m}(B, A_1, \dots, A_L)$.

Assume that the hypothesis holds for every ordered prefix of length $\ell+1$. To shorten notations, let us define $T_\ell := (B, A_1, A_2, \dots, A_\ell)$ and $T_j := (B, A_1, \dots, A_\ell, A_j)$. We then evaluate:

$$\begin{aligned} S(B, A_1, A_2, \dots, A_\ell) &= \tilde{m}(B, A_1, A_2, \dots, A_\ell) + \sum_{j=\ell+1}^L S(B, A_1, A_2, \dots, A_\ell, A_j) \\ &= \tilde{m}(T_\ell) + \sum_{j=\ell+1}^L S(T_j) \\ &\text{which by inductive hypothesis gives} \\ &= \tilde{m}(T_\ell) + \sum_{j=\ell+1}^L \tilde{m}(T_j) \times \frac{Pl_\ell(E) - m(A_j)}{Pl_L(E)} \\ &\text{here we can substitute the recursive formula for } \tilde{m}(T_j) \text{ to get} \\ &= \tilde{m}(T_\ell) + \sum_{j=\ell+1}^L \tilde{m}(T_\ell) \times \frac{m(A_j)}{Pl_\ell(E) - m(A_j)} \times \frac{Pl_\ell(E) - m(A_j)}{Pl_L(E)} \\ &= \tilde{m}(T_\ell) + \sum_{j=\ell+1}^L \tilde{m}(T_\ell) \times \frac{m(A_j)}{Pl_L(E)} \\ &= \frac{\tilde{m}(T_\ell)}{Pl_L(E)} \left[Pl_L(E) + \sum_{j=\ell+1}^L m(A_j) \right]. \end{aligned}$$

Finally, recall that we defined $m_0 = m$, and that m_k is obtained from m_{k-1} by transferring the mass of A_k entirely outside of E . Transferring the mass of A_1 then exactly drops the plausibility by $m(A_1)$, such that $Pl_1(E) = Pl_0(E) - m(A_1)$. Since the same mass transfer is performed at every iteration, we can extend the same principle

universally, so we see $Pl_\ell(E) = Pl_L(E) + \sum_{j=\ell+1}^L m(A_j)$. Substituting this equality yields

$$S(B, A_1, A_2, \dots, A_\ell) = \tilde{m}(T_\ell) \times \frac{Pl_\ell(E)}{Pl_L(E)}.$$

This shows that if our inductive hypothesis holds at order $(\ell + 1)$, then it does so too at order ℓ . Hence the explicit formulation for $S(B, A_1, A_2, \dots, A_\ell)$ holds.

In particular, for $\ell = 0$, we have the base sequence $T_0 = (B)$, which gives

$$S(B) = \tilde{m}(B) \times \frac{Pl_0(E)}{Pl_L(E)} = \frac{m(B)}{Pl_0(E)} \times \frac{Pl_0(E)}{Pl_L(E)} = \frac{m(B)}{Pl_L(E)}.$$

What then remains is to show that

$$Bel(X \mid E) = \sum_{B \in \mathcal{B}(X)} \frac{m(B)}{Pl_L(E)}.$$

Showing equivalence to the FH closed form

To complete the proof, we must find equivalence between

$$\frac{\sum_{B \in \mathcal{B}(X)} m(B)}{Pl_L(E)},$$

and the FH closed form;

$$\frac{Bel(X)}{Bel(X) + Pl_0(E \setminus X)}.$$

It is straightforward that the numerator $\sum_{B \in \mathcal{B}(X)} m(B)$ is equal to $Bel(X)$, since by definition $\mathcal{B}(X)$ contains all focal elements completely contained in X ;

$$\sum_{B \in \mathcal{B}(X)} m(B) = Bel(X).$$

For the denominator, let us evaluate the plausibility of E at iteration L . Recall that $Pl_L(E) = Pl_0(E) - \sum_{j=1}^L m(A_j)$. By definition, $Pl_L(E) = 1 - Bel(E^c) - \sum_{j=1}^L m(A_j)$. Furthermore, the plausibility of the space outside X is $Pl_0(E \setminus X) = 1 - Bel(E^c \cup X) = 1 - Bel(E^c) - Bel(X) - \sum_{j=1}^L m(A_j)$.

Isolating the terms then shows $Pl_L(E) - Pl_0(E \setminus X) = Bel(X)$, which rearranges to $Pl_L(E) = Bel(X) + Pl_0(E \setminus X)$, giving us the correct denominator.

This demonstrates that m' is the Möbius transform of $Bel(\cdot \mid E)$. Because m' is built purely from nonnegative terms in the recurrence, its nonnegativity is self-evident. Therefore, m' is a valid basic belief assignment, and $Bel(\cdot \mid E)$ is a valid belief function.